

AI/BigData 인텐시브 과정

강사장철원

Section 0
코스 소개

DAY1	DAY2	DAY3	DAY4	DAY5
데이터 분석을 위한 기초 수학			데이터 집계 및 시각화	
DAY6	DAY7	DAY8	DAY9	DAY10
	미니 프로젝트	확률분포, AB테스트		
DAY11	DAY12	DAY13	DAY14	DAY15
머신러닝				
DAY16	DAY17	DAY18	DAY19	DAY20
딥러닝		파이널 프로젝트		

□ 딥러닝의 개념

□ 퍼셉트론 기초

□ 신경망의 개념

AI/BigData 인텐시브 과정

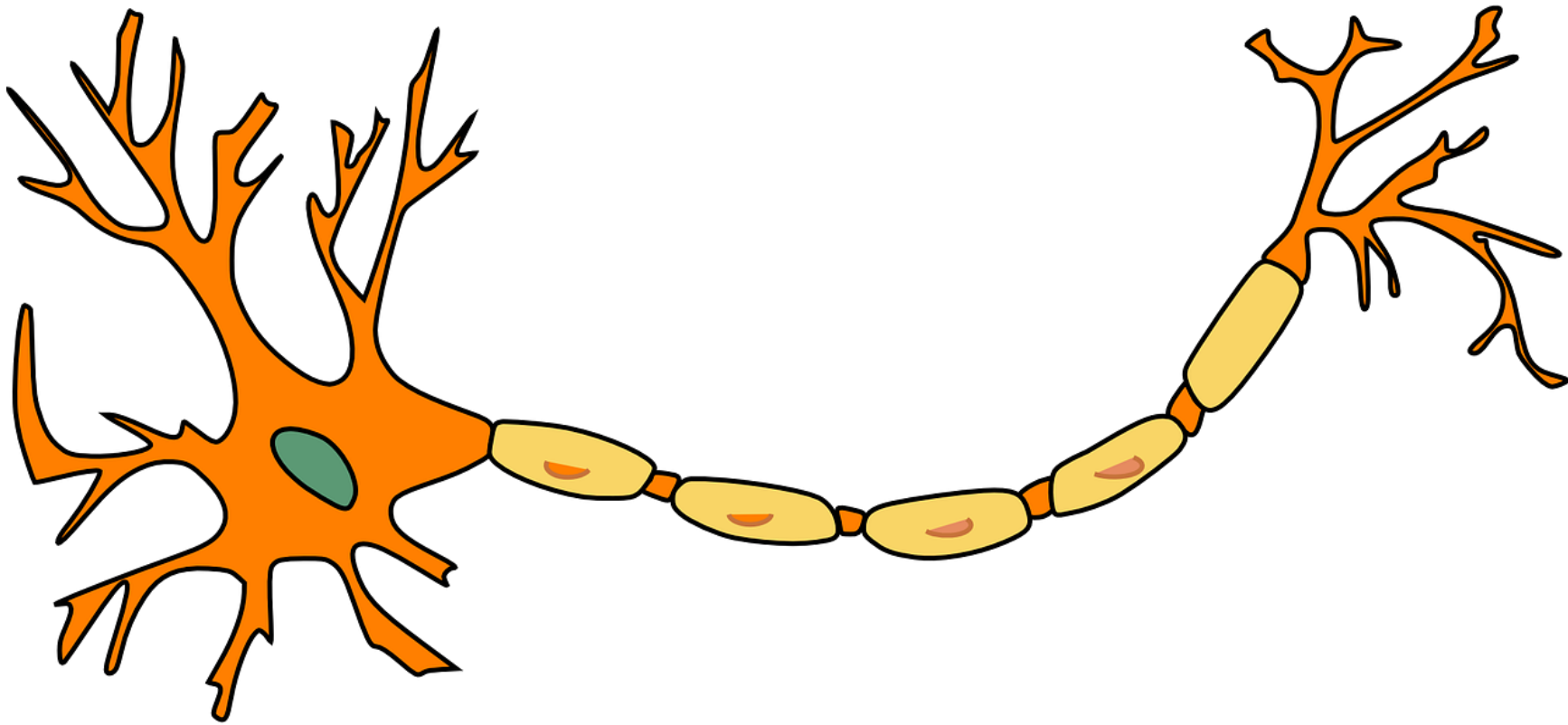
Section 1. 딥러닝 기초

Section 1-1. 딥러닝의 개념

인공 신경망(Artificial Neural Network)

- 인간의 뇌는 뉴런이라는 세포로 구성되어 있음(약 1,000억개)
- 뉴런은 시냅스로 연결되어 있음
- 자극의 크기가 특정 값이상이면 뉴런에 전달
- 인공 신경망은 이와 같은 아이디어를 머신 러닝에 적용한 것

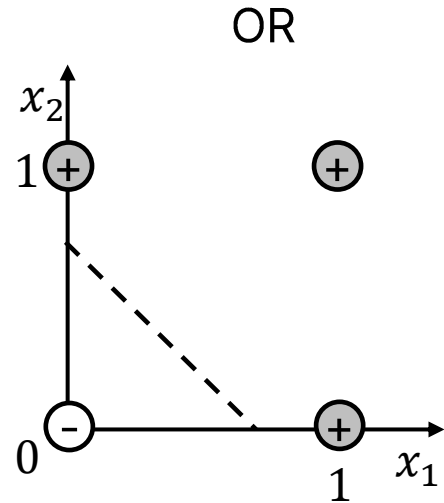
뉴런(neuron)



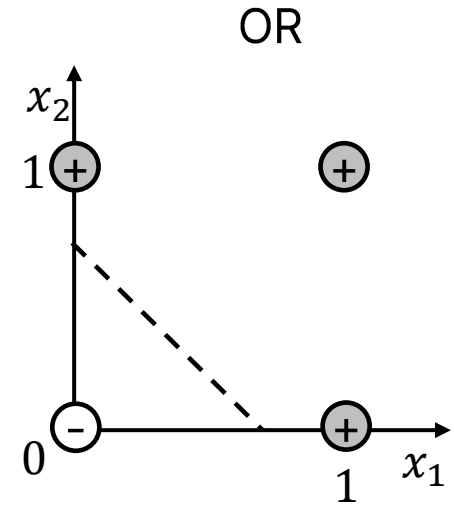
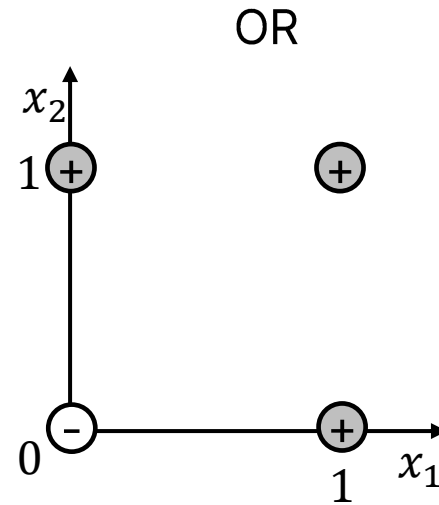
딥러닝(deep learning)

- 신경망을 기반으로 학습하는 방법
- 딥러닝에는 수많은 퍼셉트론이 연결되어 연산 결과를 주고 받음

초기 딥러닝 문제: XOR 문제

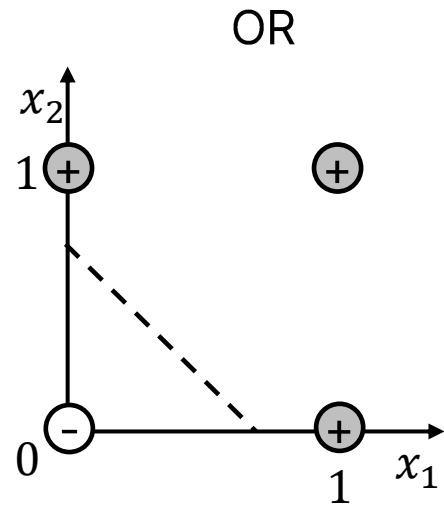


x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

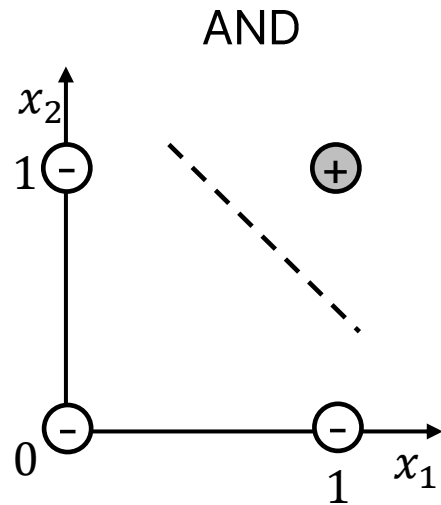


직선 하나로 0과 1값을 구분할 수 있음

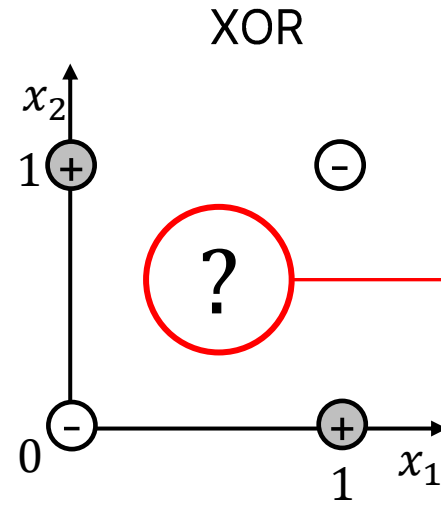
초기 딥러닝 문제: XOR 문제



x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1



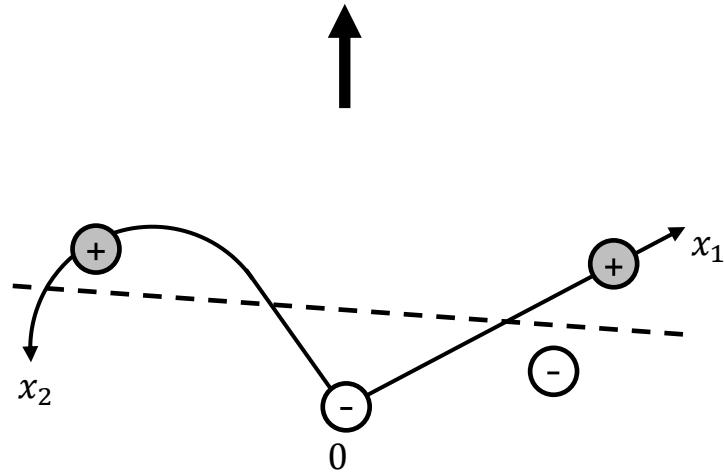
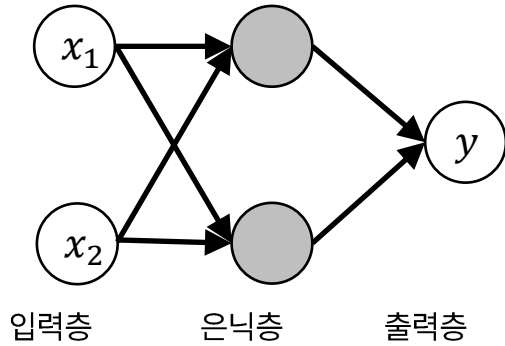
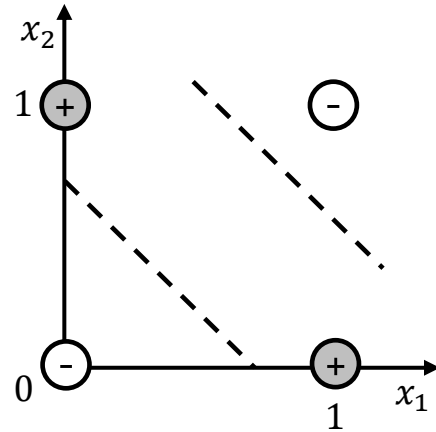
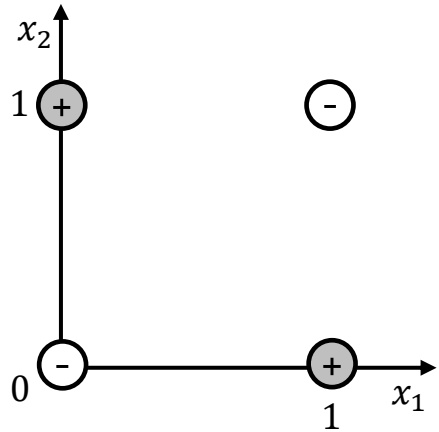
x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1



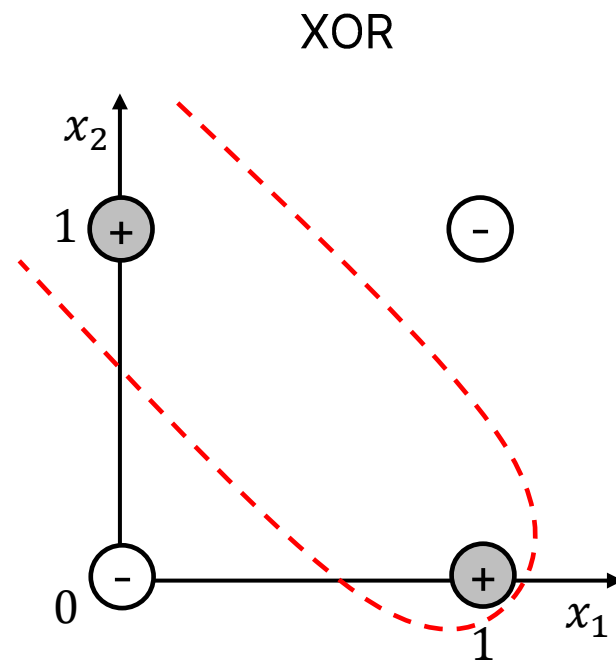
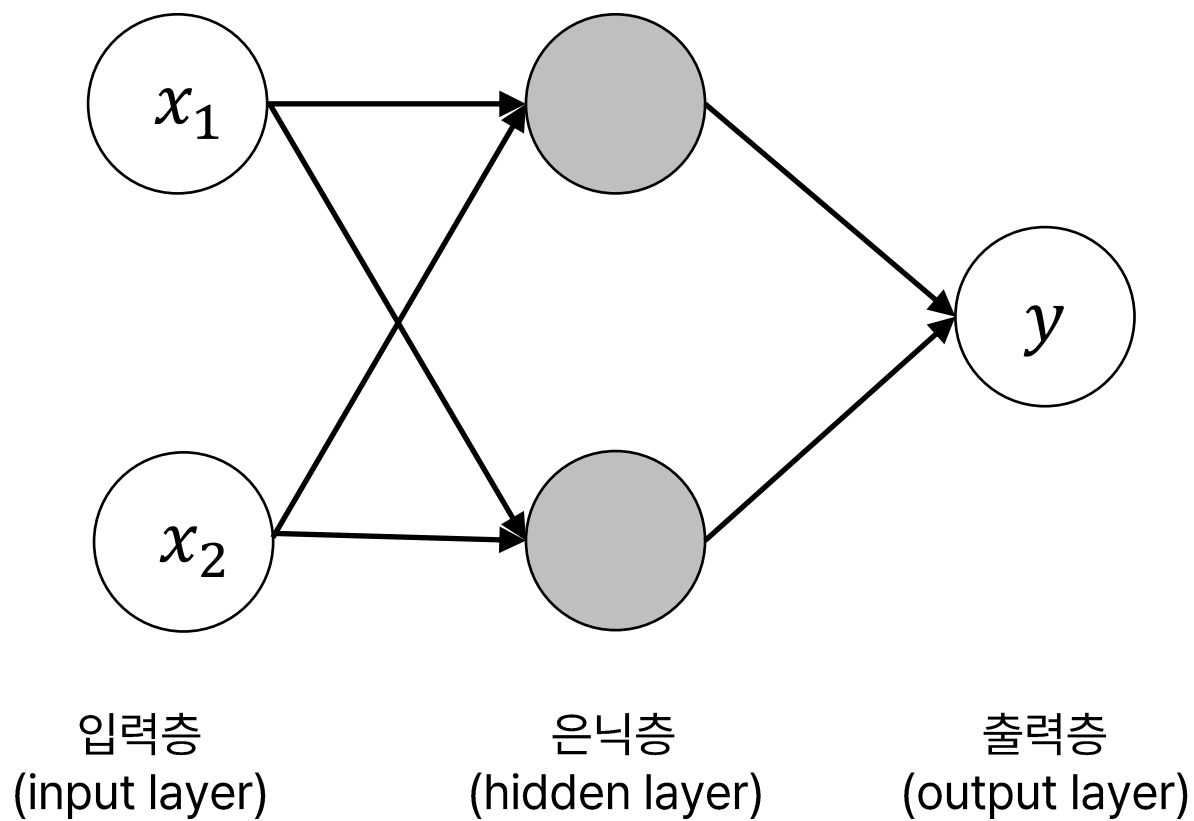
x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

직선 하나로 0과 1값을 구분
할 수 없음

초기 딥러닝 문제: XOR 문제

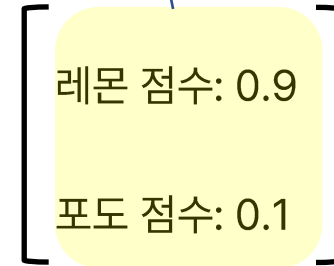
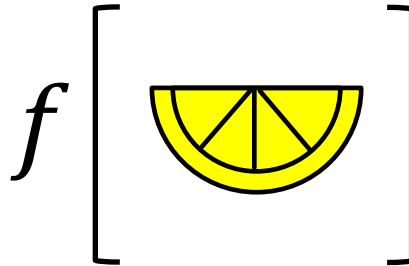
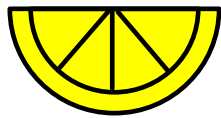


초기 딥러닝 문제: XOR 문제

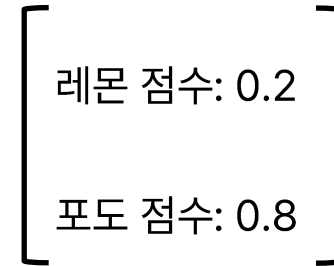
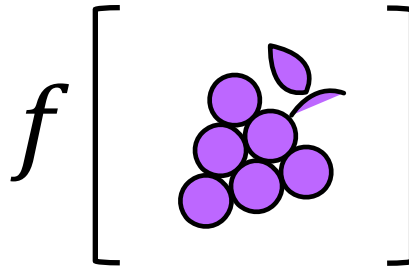
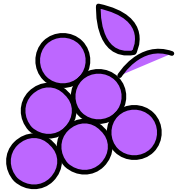


딥러닝의 기초 개념

f : 레몬, 포도 분류기



레몬



포도

레몬과 포도에 대한 판
별 점수 출력



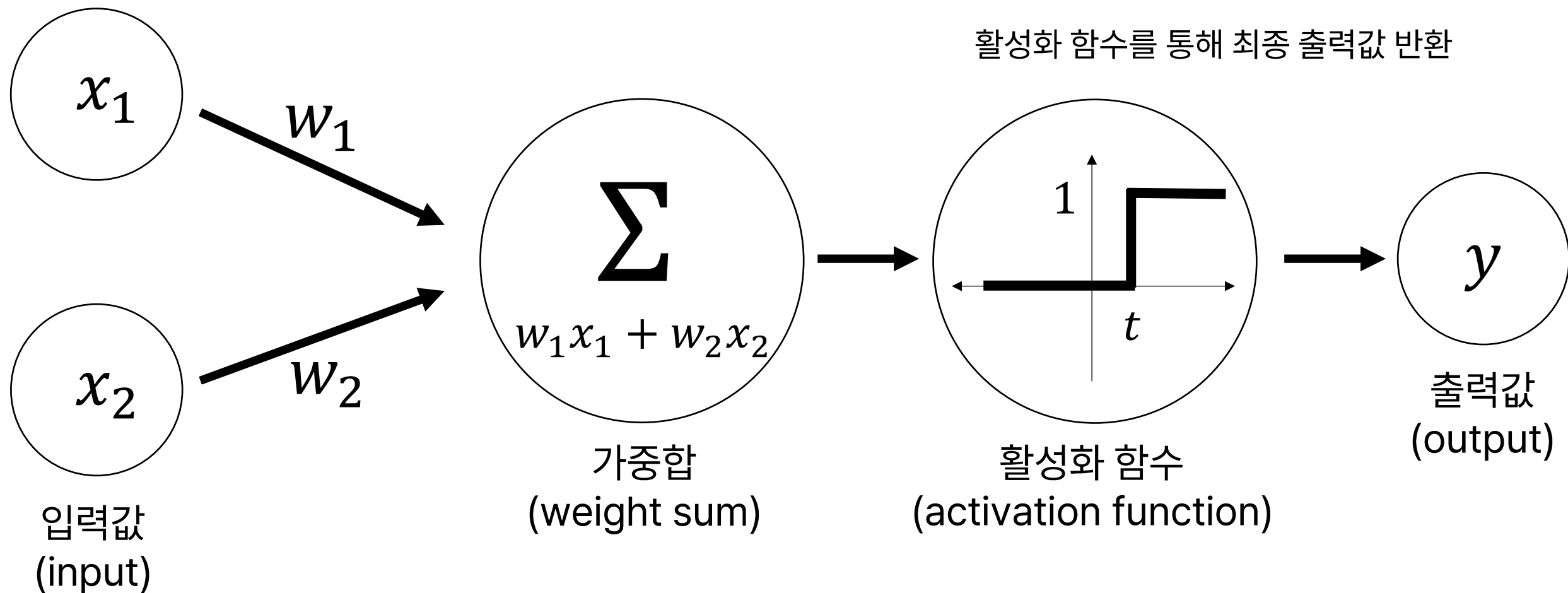
이미지 데이터 입력

AI/BigData 인텐시브 과정

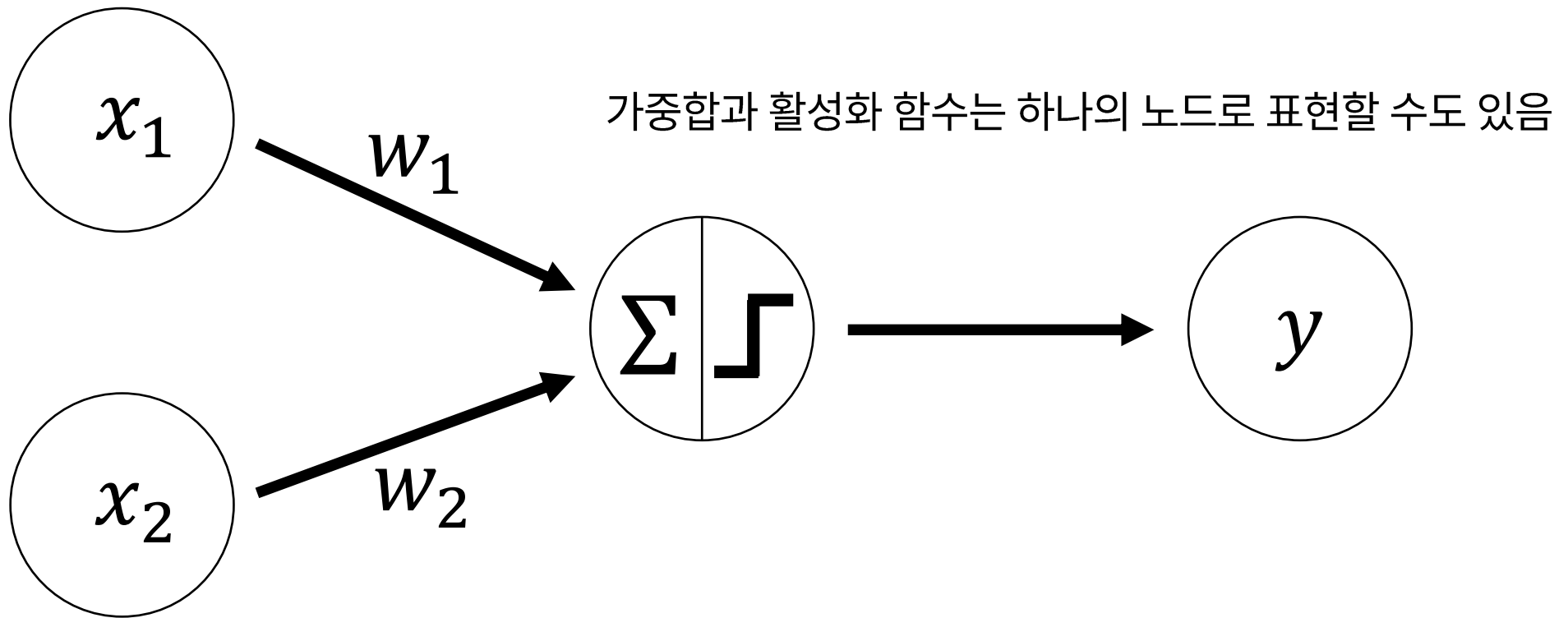
Section 1. 딥러닝 기초

Section 1-2. 퍼셉트론 기초

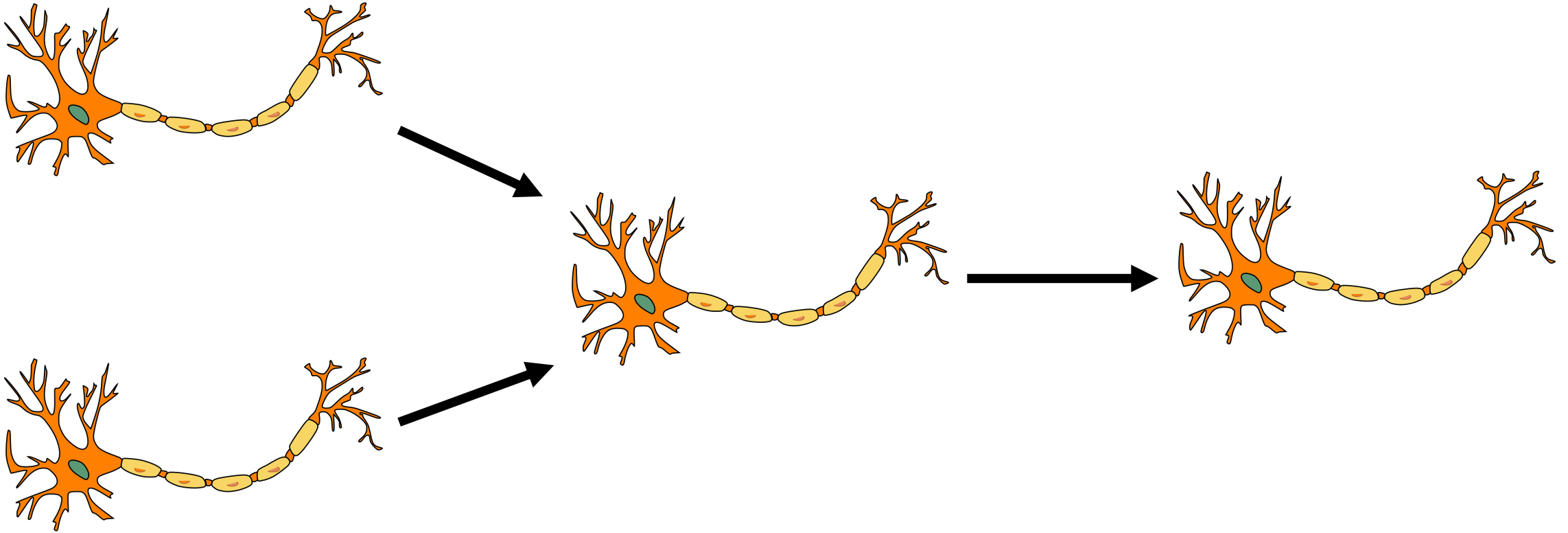
퍼셉트론 구조



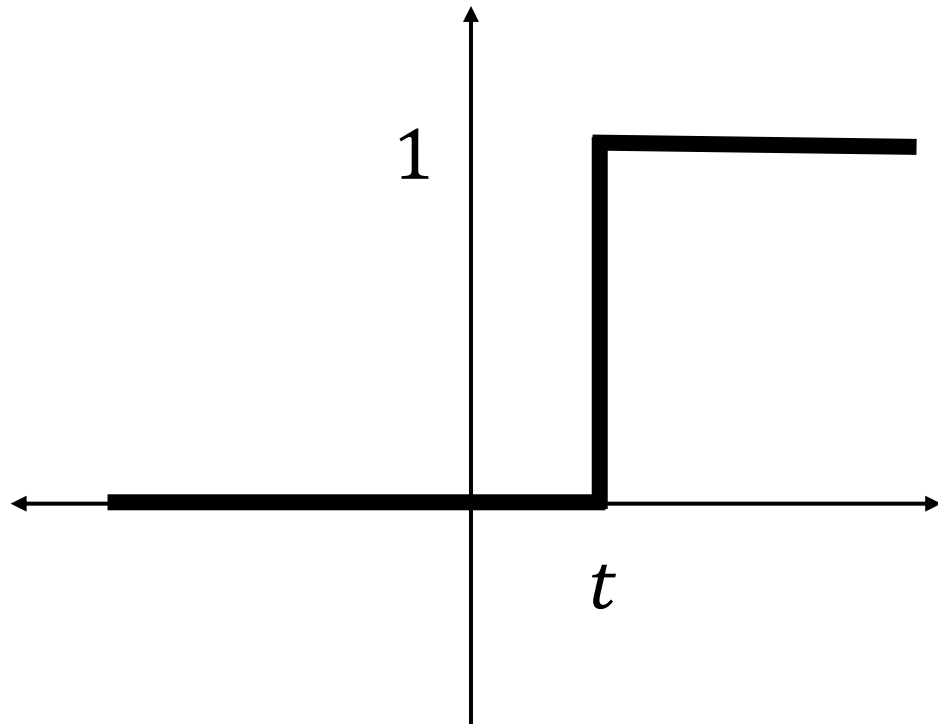
퍼셉트론 구조



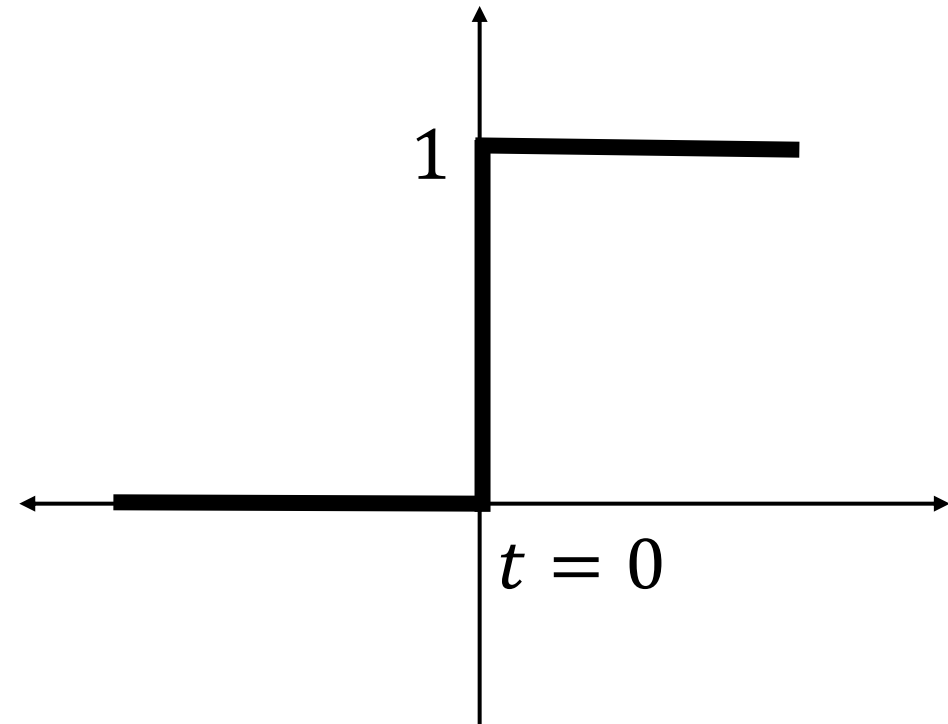
노드(node) = 뉴런(neuron)



퍼셉트론 구조

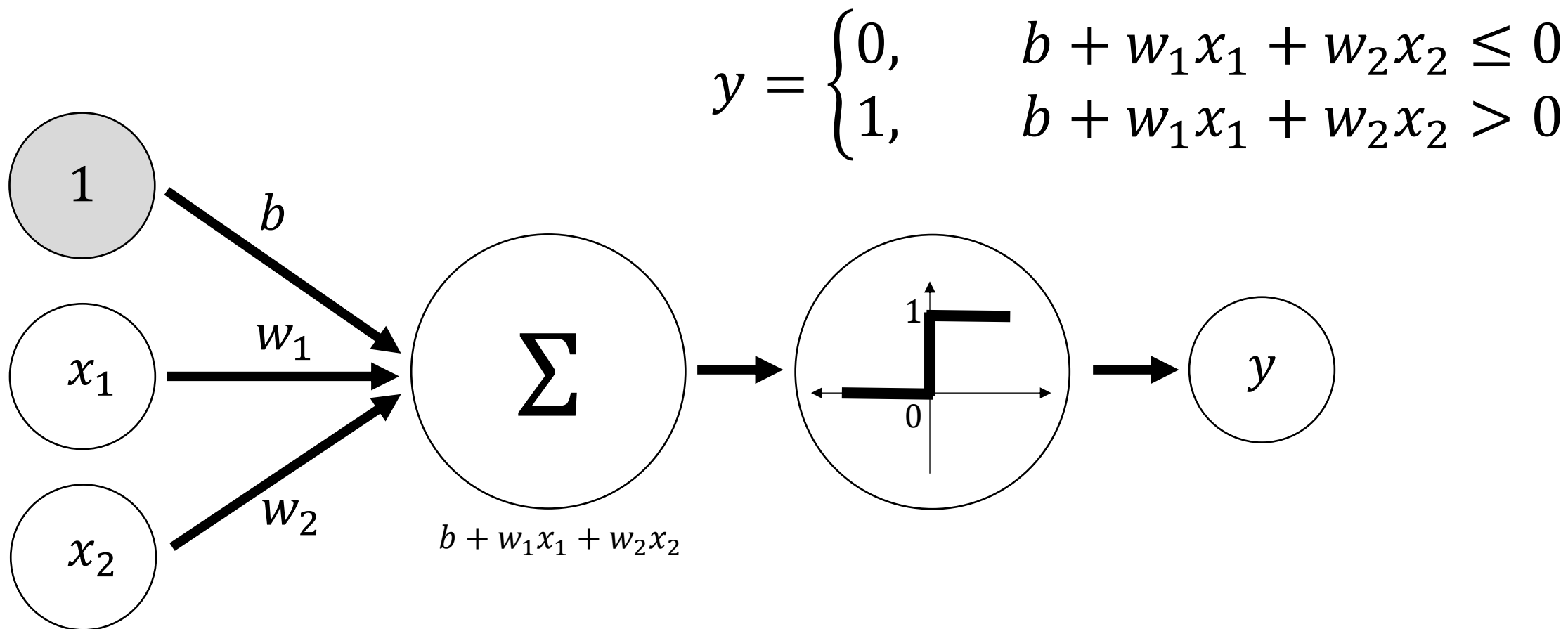


임계치 = t

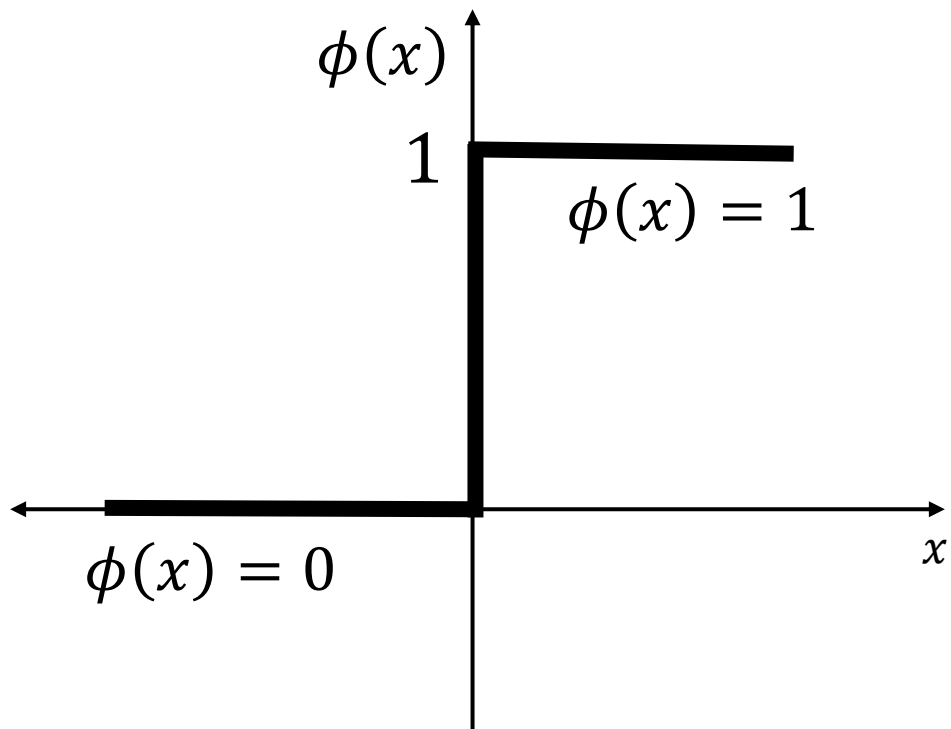


임계치 = 0

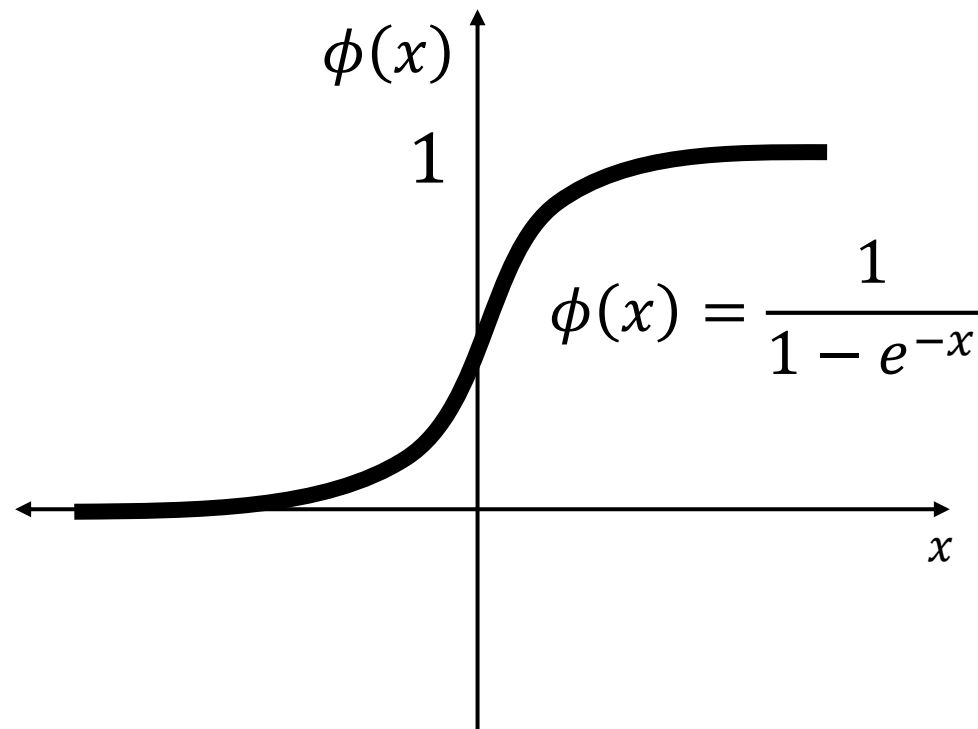
퍼셉트론 구조



퍼셉트론 구조

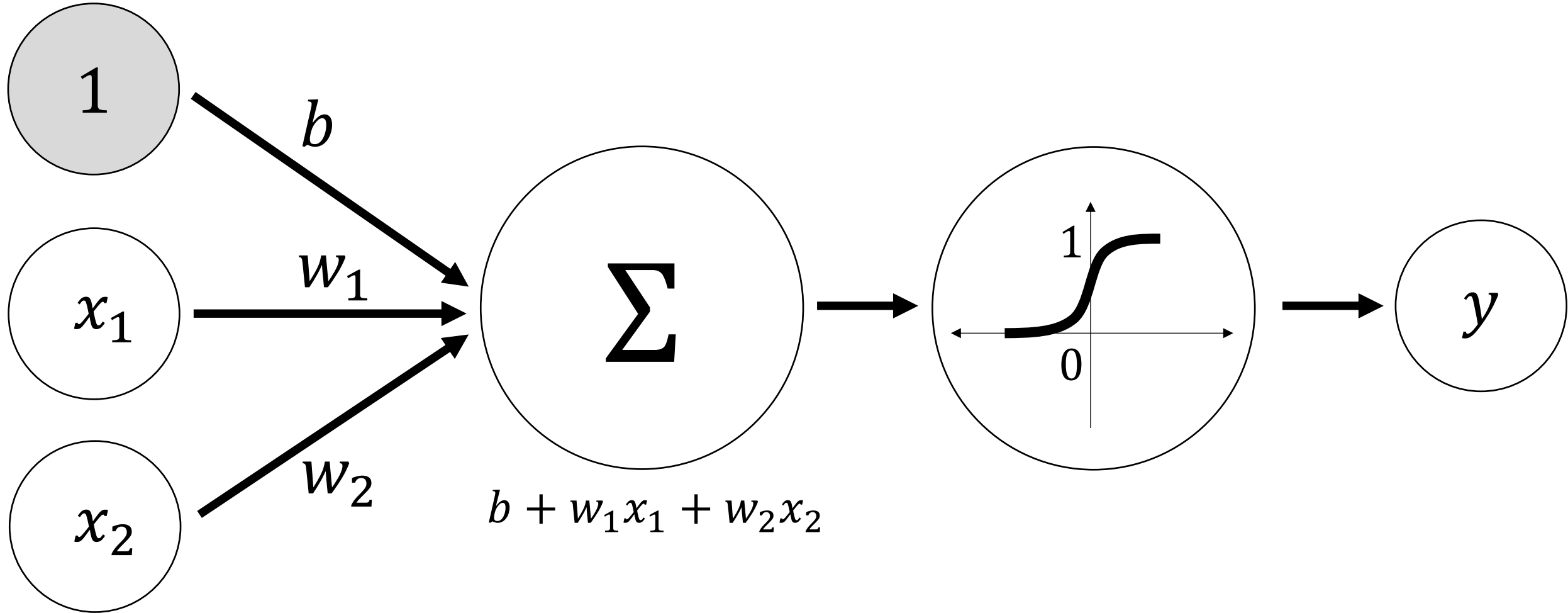


계단 함수
(step function)



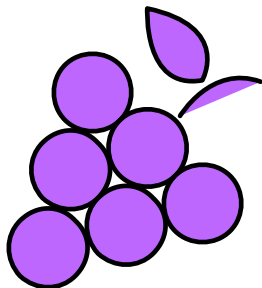
시그모이드 함수
(sigmoid function)

활성화 함수를 시그모이드 함수로 변경



퍼셉트론으로 분류하기

실제 이미지



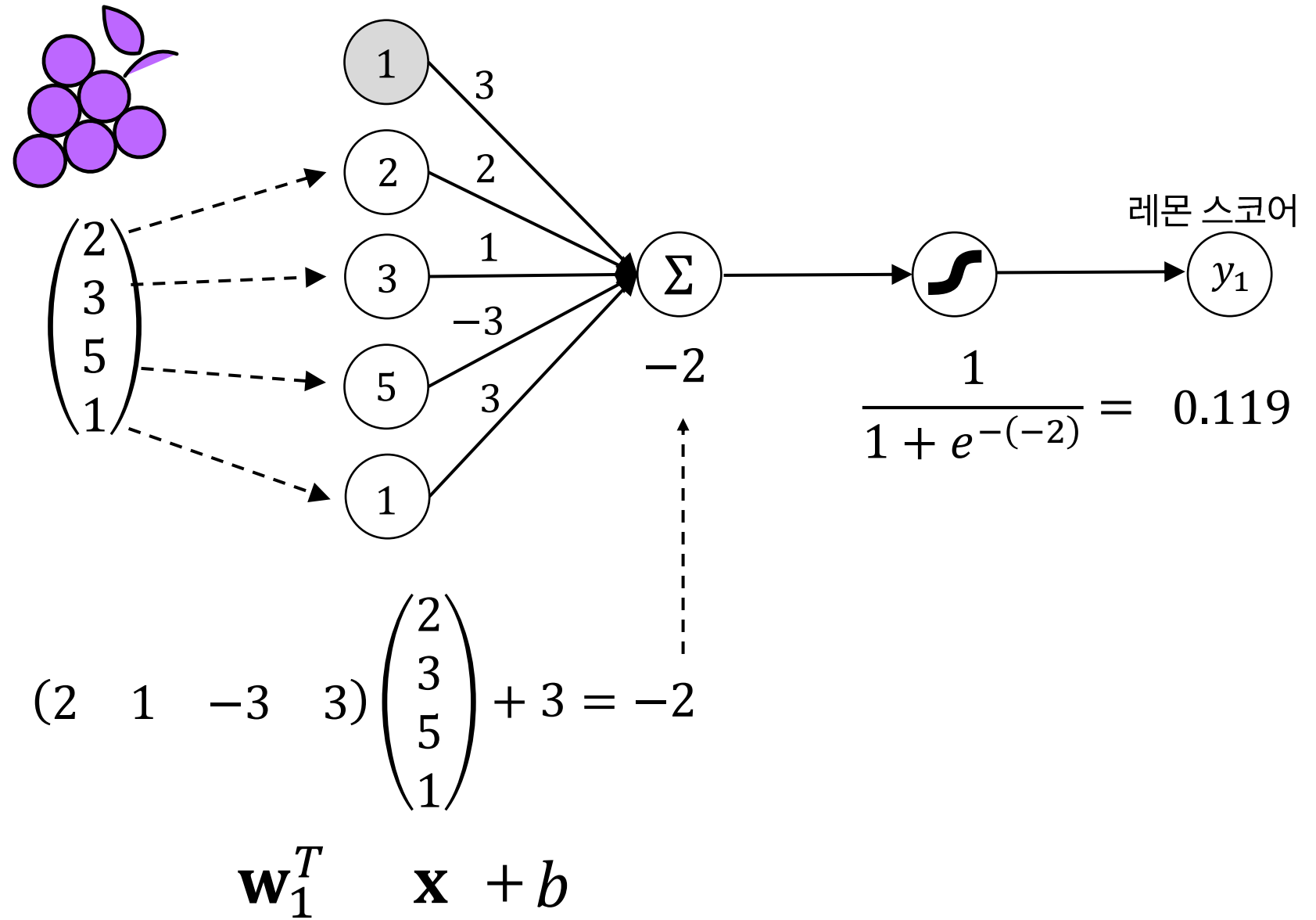
이미지 데이터 행렬

$$\begin{pmatrix} 2 & 3 \\ 5 & 1 \end{pmatrix}$$

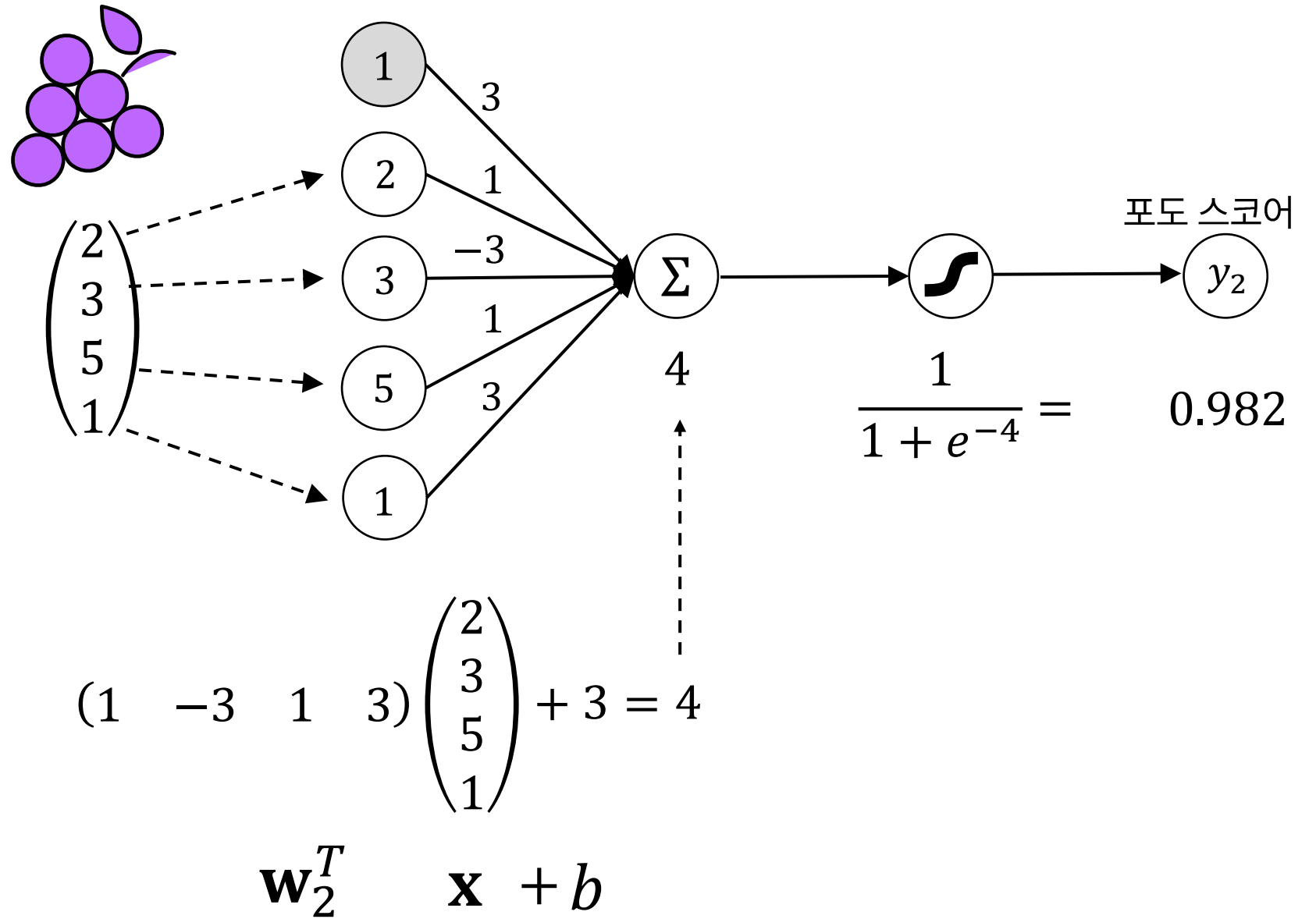
이미지 데이터 벡터

$$\begin{pmatrix} 2 \\ 3 \\ 5 \\ 1 \end{pmatrix}$$

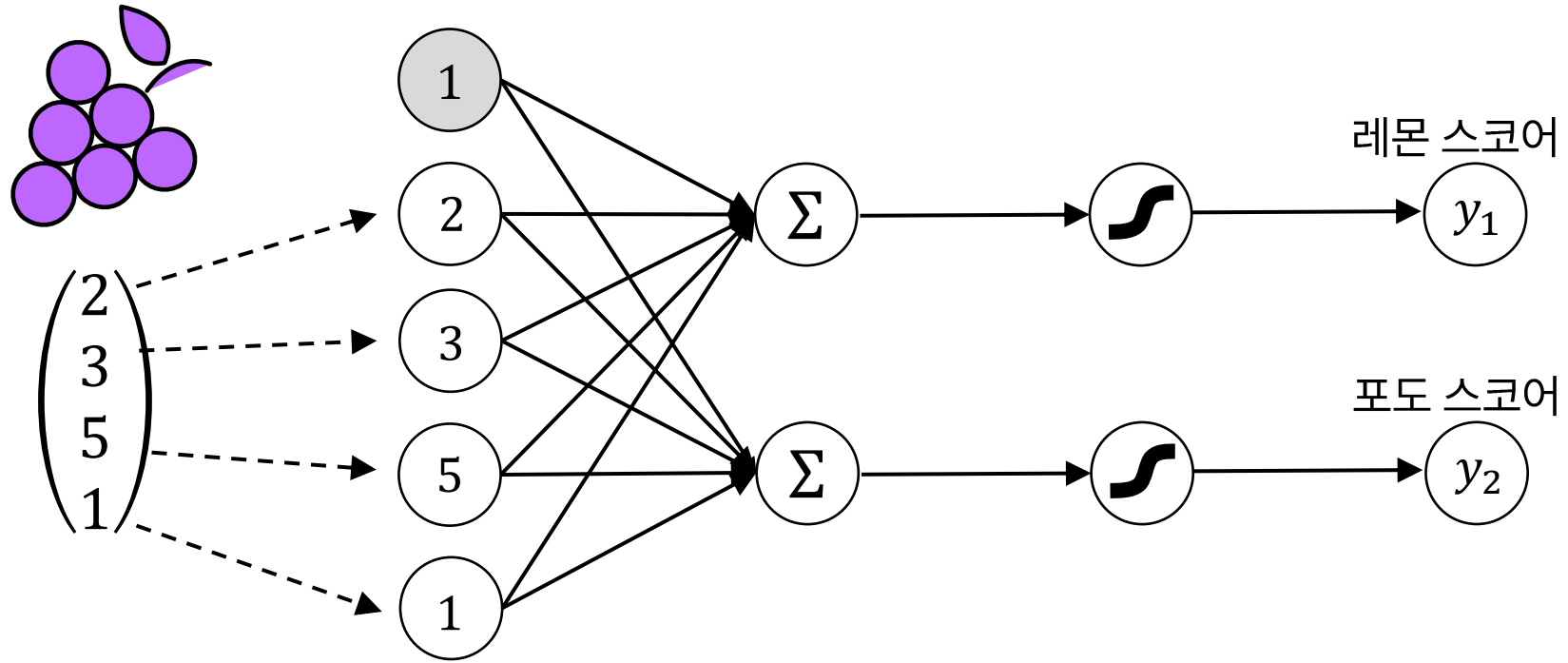
퍼셉트론 분류



퍼셉트론 분류



Section
퍼셉트론기초



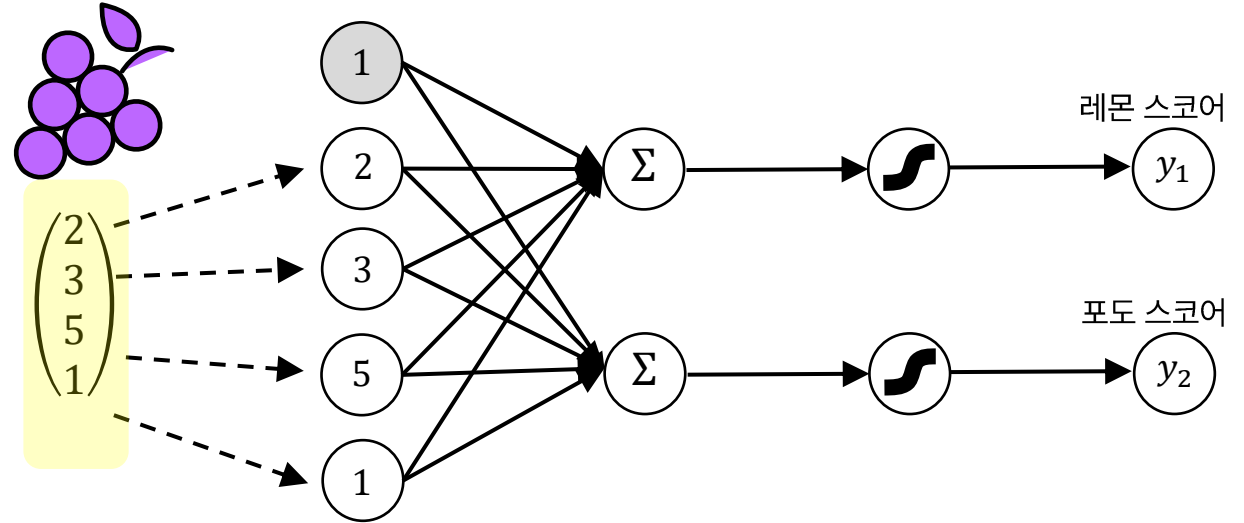
$$\begin{pmatrix} 2 & 1 & -3 & 3 \\ 1 & -3 & 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \xrightarrow{\frac{1}{1+e^{-x}}} \begin{pmatrix} 0.119 \\ 0.982 \end{pmatrix} \longrightarrow \text{포도}$$

$\mathbf{W}^T \mathbf{x} + \mathbf{b}$

Section 퍼셉트론기초

```
1 import numpy as np
2
3 # 입력층
4 input_data = np.array([[2,3], [5,1]])
5 print(input_data)
6 x = input_data.reshape(-1)
7 print(x)
```

```
[[2 3]
 [5 1]]
[2 3 5 1]
```



$$\begin{pmatrix} 2 & 1 & -3 & 3 \\ 1 & -3 & 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \xrightarrow{\frac{1}{1+e^{-x}}} \begin{pmatrix} 0.119 \\ 0.982 \end{pmatrix} \longrightarrow \text{포도}$$

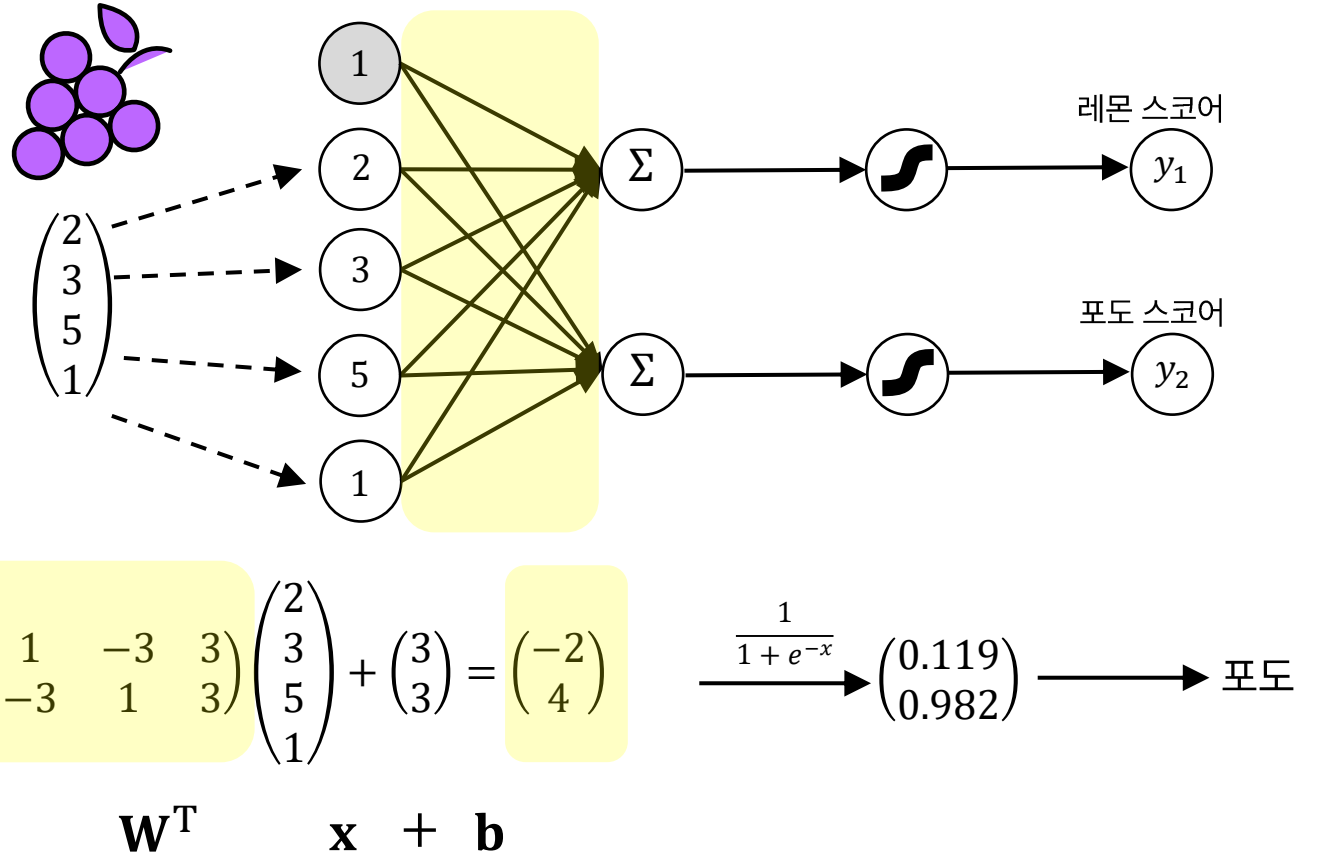
$\mathbf{W}^T \quad \mathbf{x} + \mathbf{b}$

Section 퍼셉트론기초

```
1 # 가중치 및 편향
2 w1 = np.array([2, 1, -3, 3])
3 w2 = np.array([1, -3, 1, 3])
4 b1 = 3
5 b2 = 3
```

```
1 # 가중합
2 W = np.array([w1, w2])
3 print(W)
4 b = np.array([b1, b2])
5 print(b)
6 weight_sum = np.dot(W, x) + b
7 print(weight_sum)
```

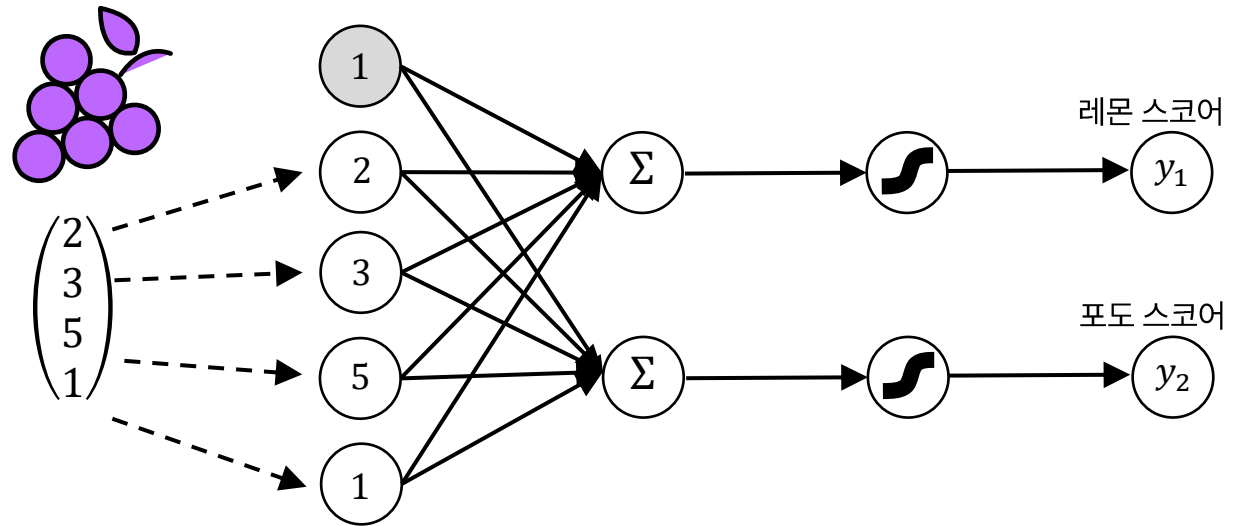
```
[[ 2  1 -3  3]
 [ 1 -3  1  3]]
[3 3]
[-2  4]
```



Section 퍼셉트론기초

```
In [5]: 1 # 출력층  
2 res = 1/(1+np.exp(-weight_sum))  
3 print(res)
```

[0.11920292 0.98201379]



$$\begin{pmatrix} 2 & 1 & -3 & 3 \\ 1 & -3 & 1 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ 5 \\ 1 \end{pmatrix} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix} \xrightarrow{\frac{1}{1+e^{-x}}} \begin{pmatrix} 0.119 \\ 0.982 \end{pmatrix} \rightarrow \text{포도}$$

$\mathbf{W}^T \mathbf{x} + \mathbf{b}$

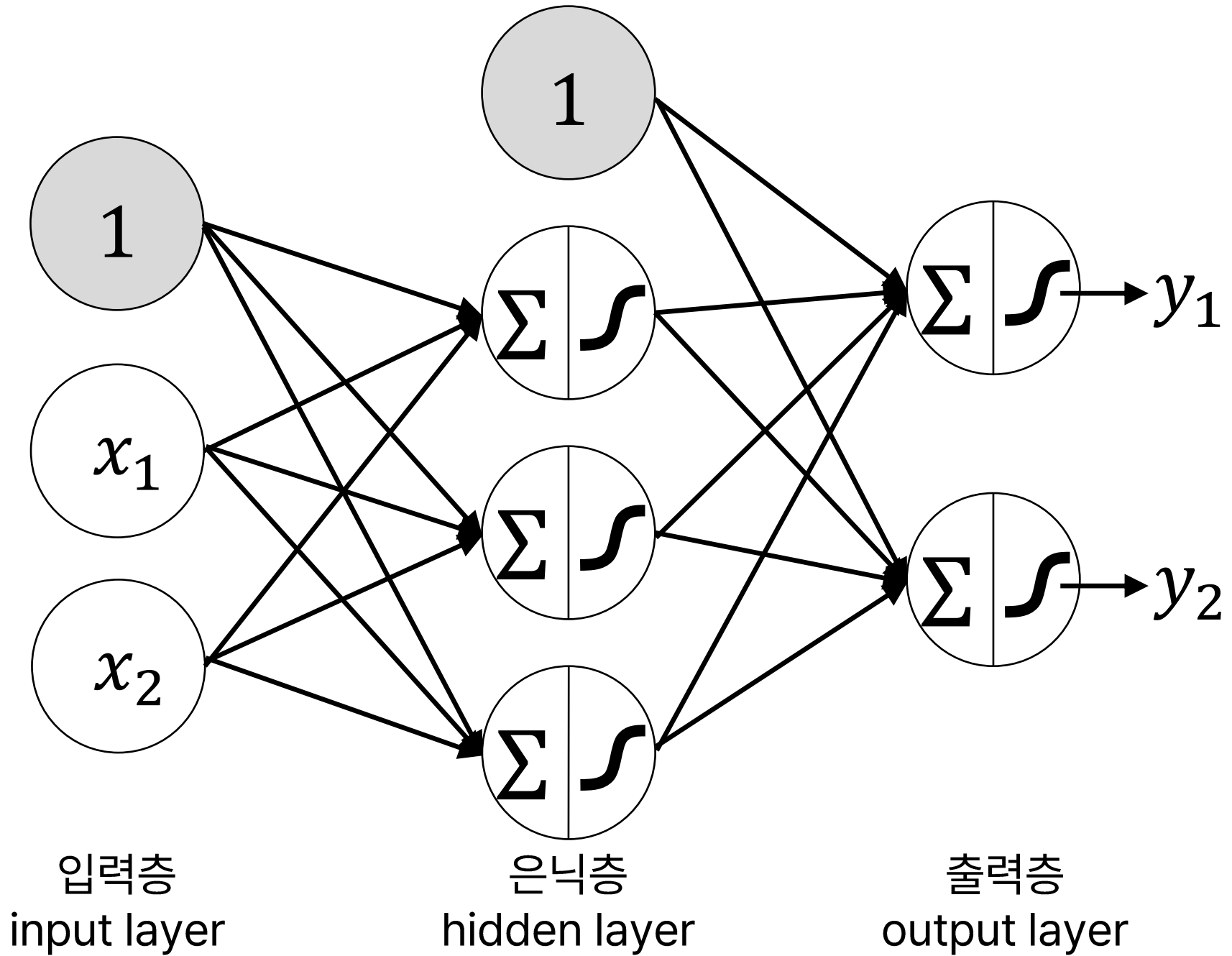
AI/BigData 인텐시브 과정

Section 1. 딥러닝 기초

Section 1-3. 신경망의 개념

신경망(neural network)

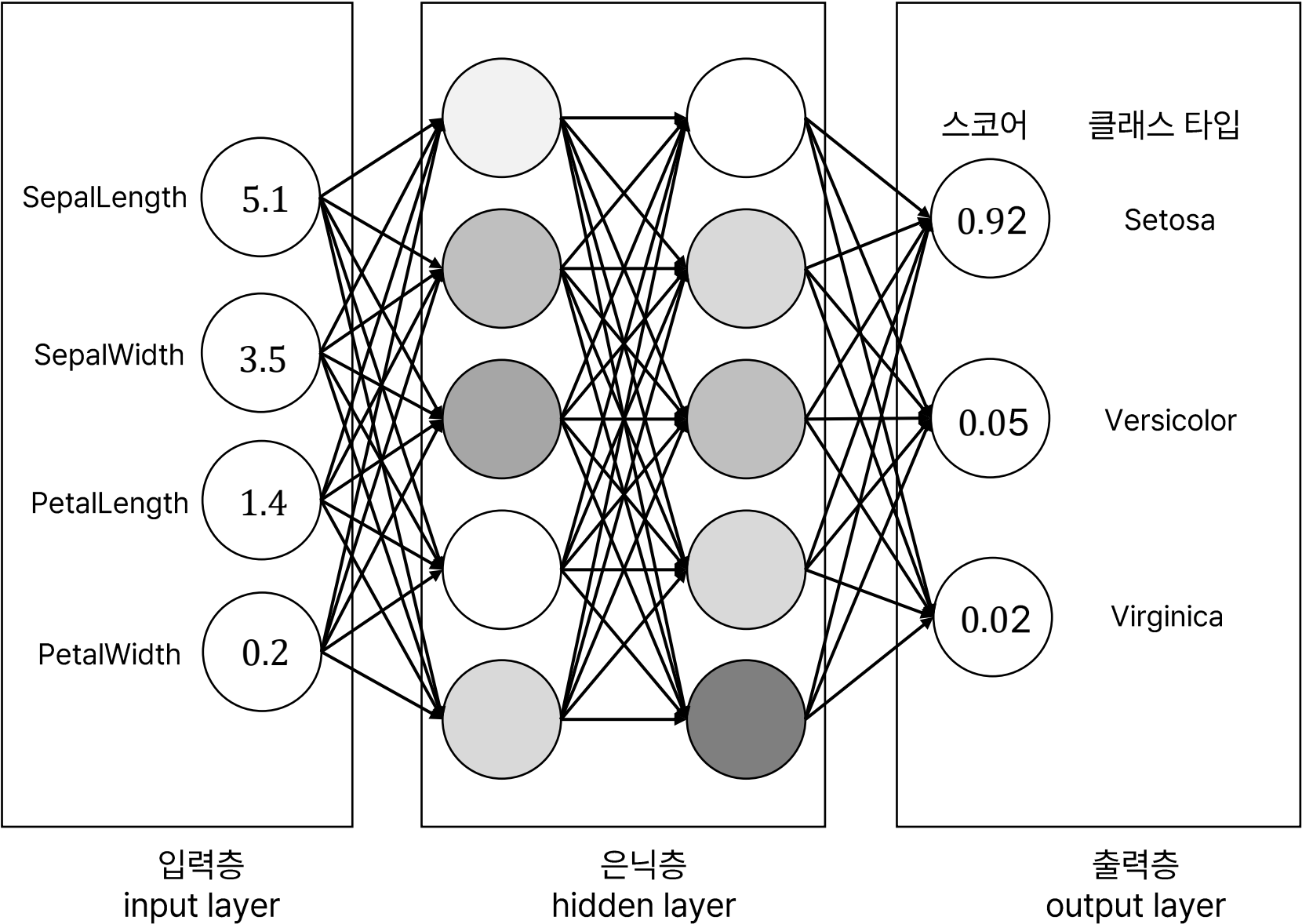
- 하나의 퍼셉트론으로 해결할 수 없는 문제를 다수의 퍼셉트론으로 해결 가능
- 다수의 퍼셉트론을 사용하는 것을 다층 퍼셉트론이라고 함
- 다수의 뉴런을 사용해 만든 것을 신경망(neural network)이라고 함



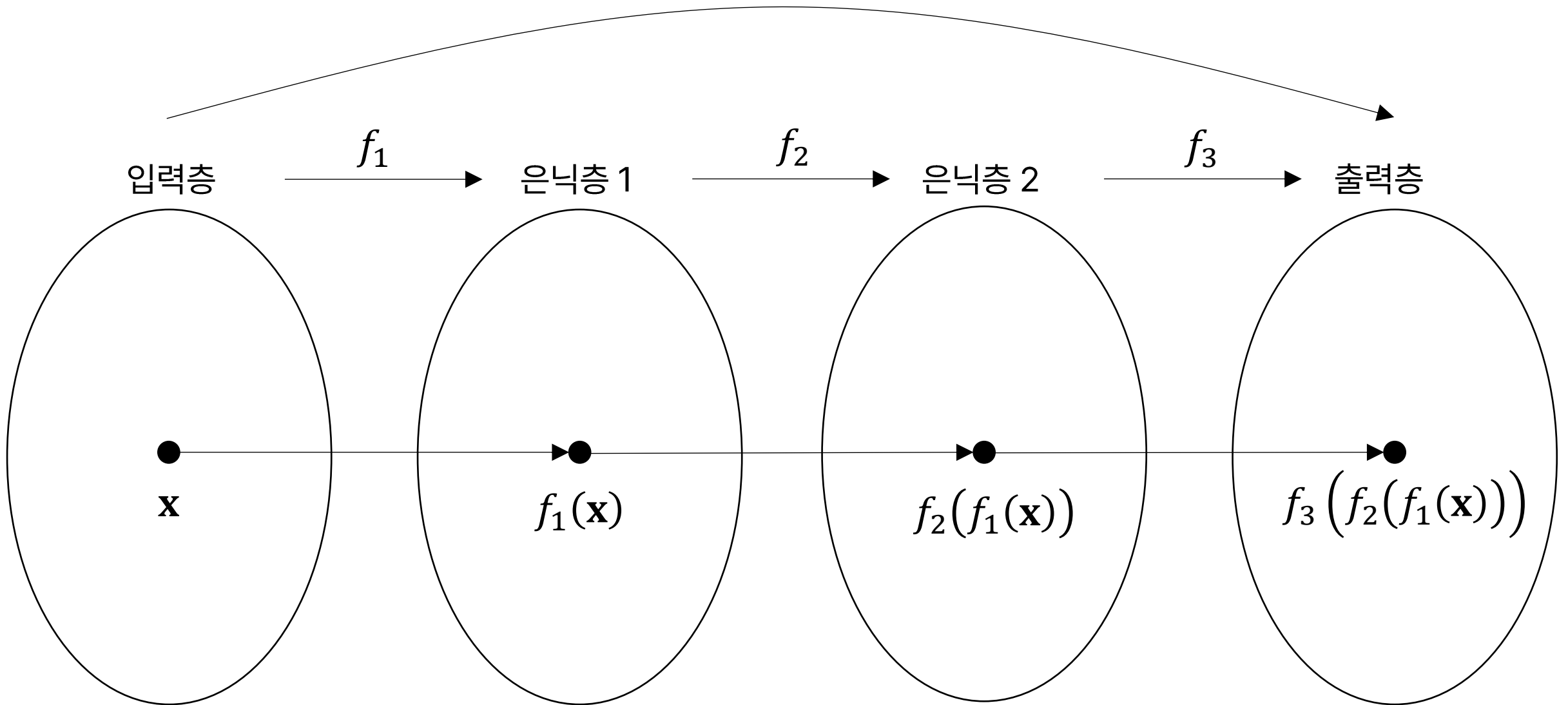
Section
신경망의개념

붓꽃 데이터

Sepal Length	Sepal Width	Petal Length	Petal Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
...	...	...	...
6.5	3.0	5.2	2.0
6.2	3.4	5.4	2.3
5.9	3.0	5.1	1.8

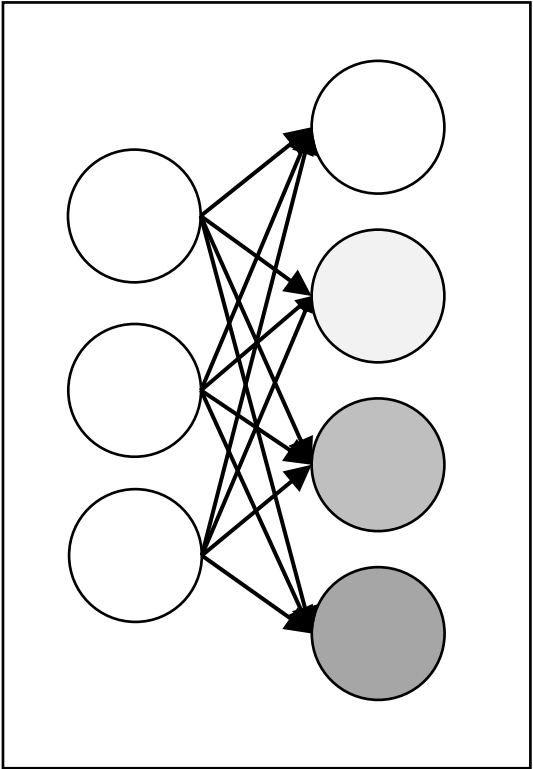


$$f(\mathbf{x}) = f_3(f_2(f_1(\mathbf{x})))$$

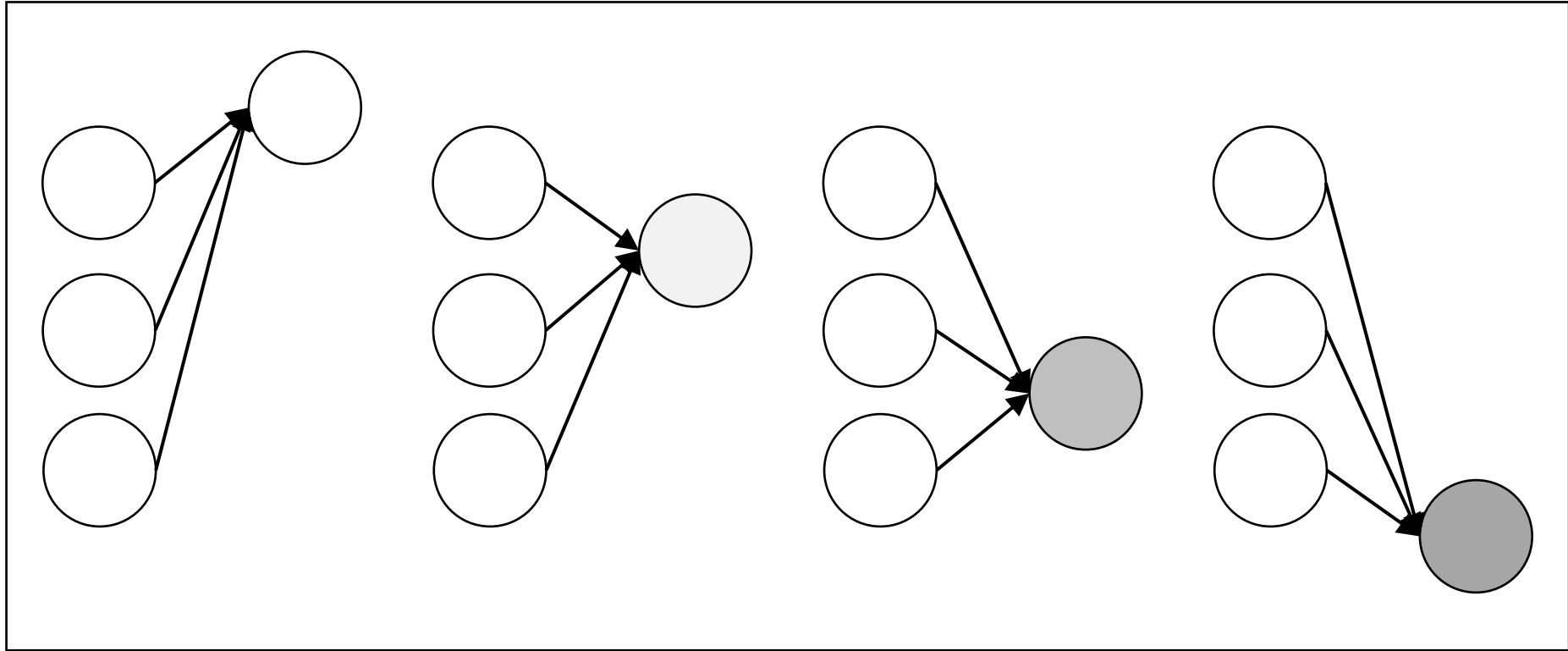


신경망(neural network)

vector-to-vector

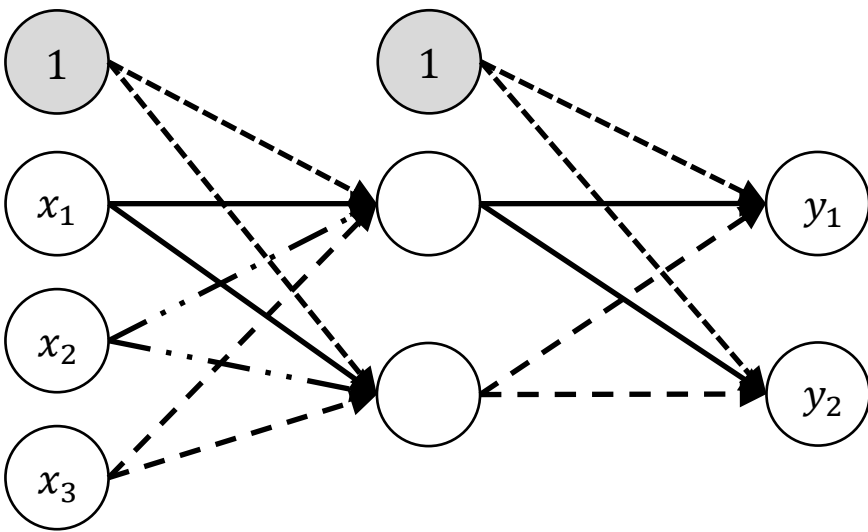


vector-to-scalar

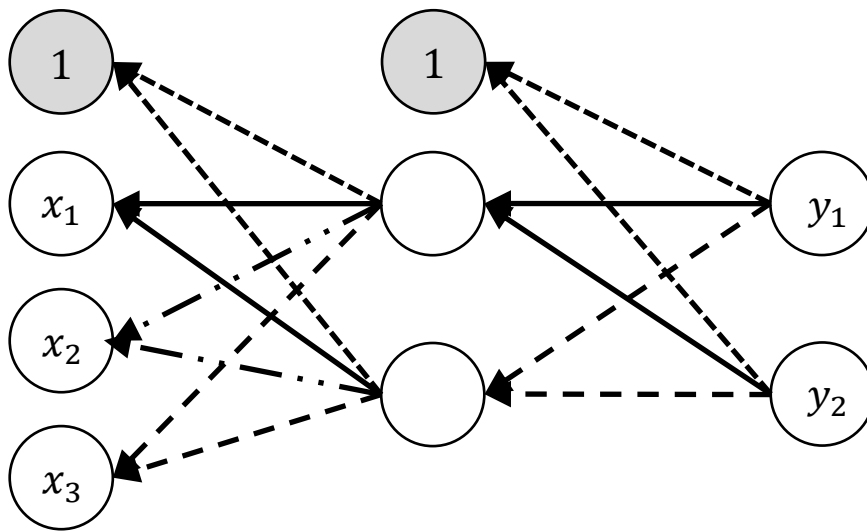


순전파와 역전파

순전파
forward propagation



역전파
back propagation



오차 역전파(backpropagation)

- (1) 가중치 초기화
- (2) 순전파(forward propagation)를 통한 출력값 계산
- (3) 비용 함수(cost function) 정의 및 1차 미분식 구하기
- (4) 역전파(back propagation)를 통한 1차 미분값 구하기
- (5) 파라미터(parameter) 업데이트
- (6) (2)~(5) 과정 반복

오차 역전파 예제

입력값: $x_1 = 1, \quad x_2 = 2, \quad x_3 = 3$

타겟: $t_1 = 0.5, \quad t_2 = 0.1$

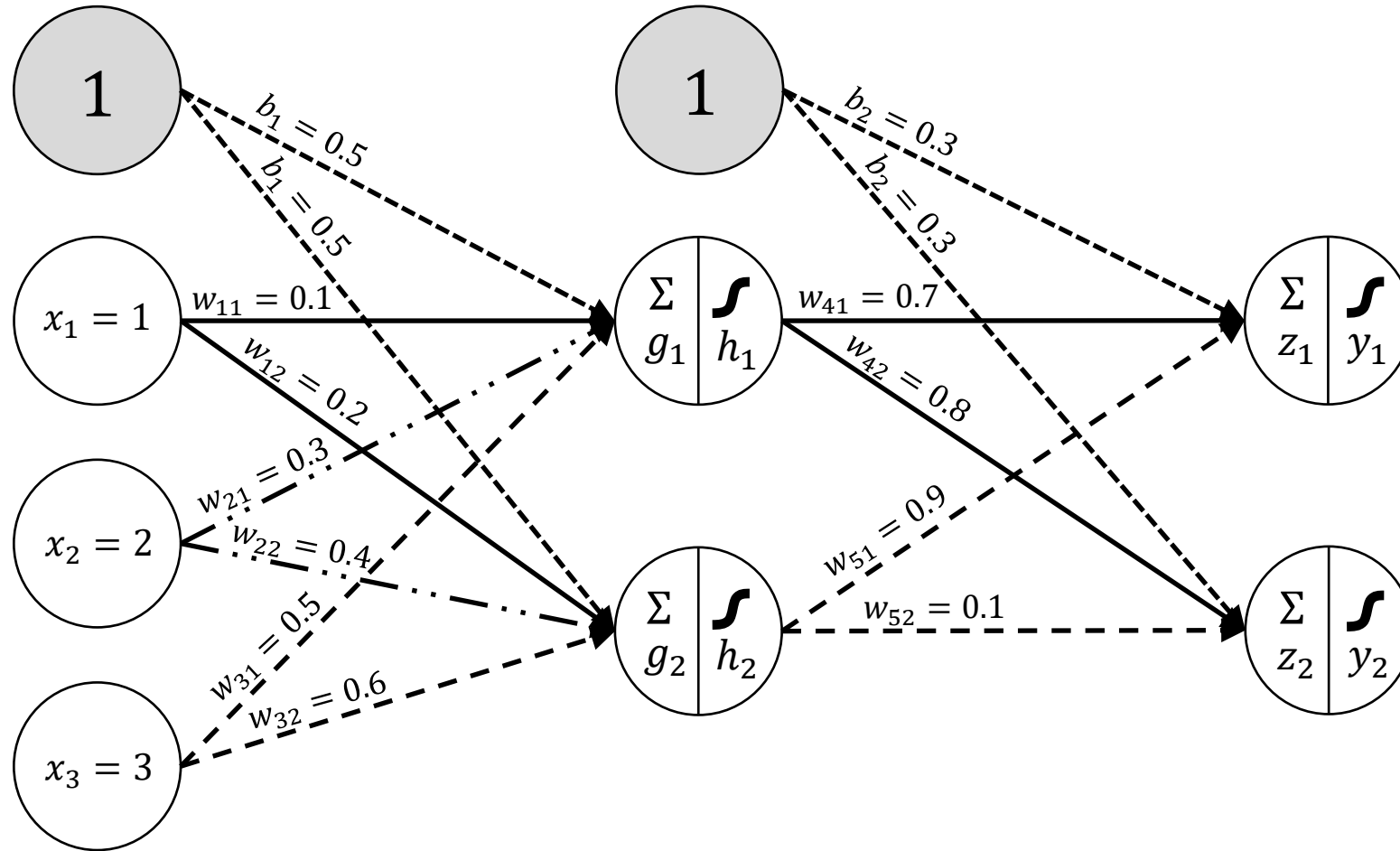
활성화 함수: 시그모이드 함수

가중치 초기화 방법: 랜덤 선택

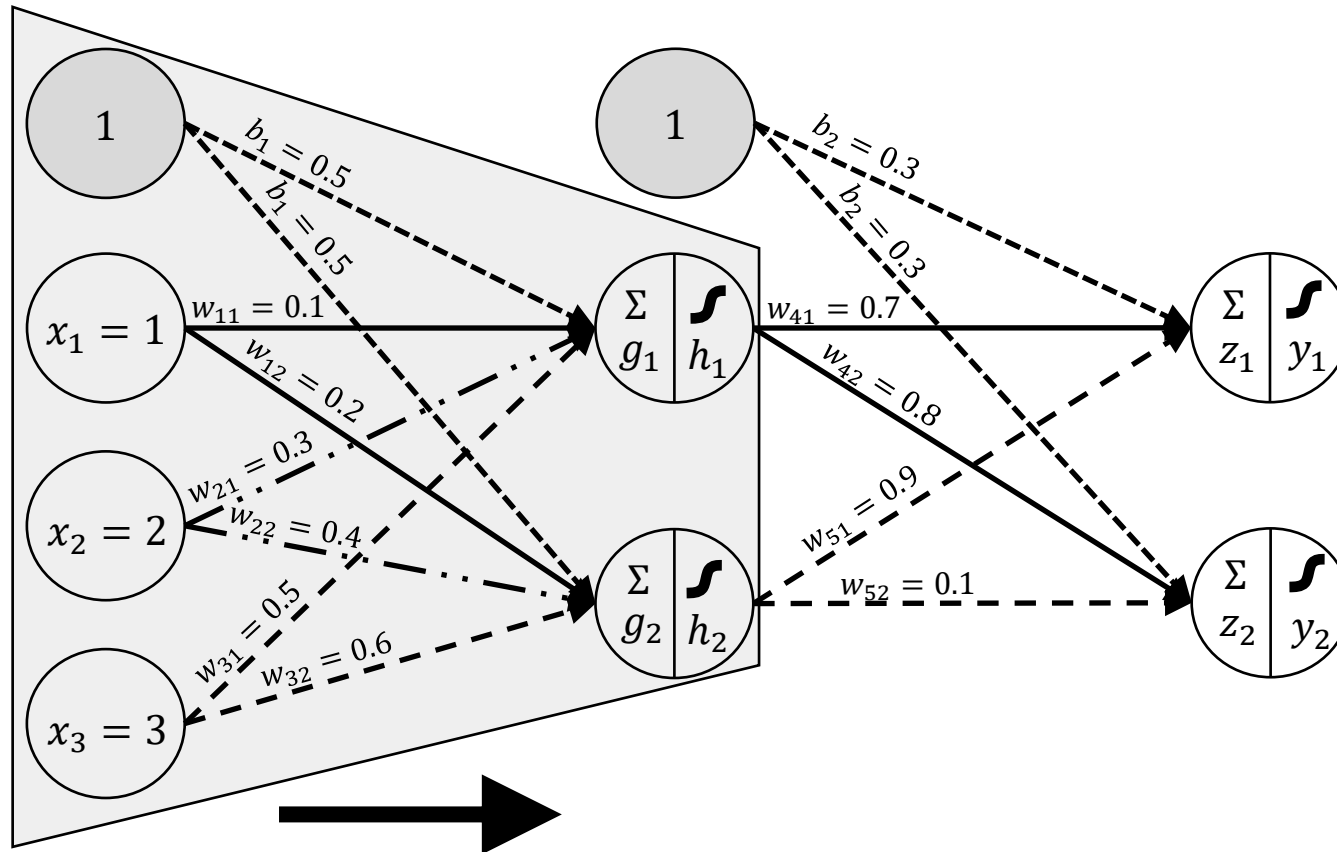
비용 함수: 오차 제곱합

학습율: 0.01

(1) 가중치 초기화



(2) 순전파를 이용한 출력값 계산



(2) 순전파를 이용한 출력값 계산

$$\mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1 = \begin{pmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_1 \end{pmatrix}$$

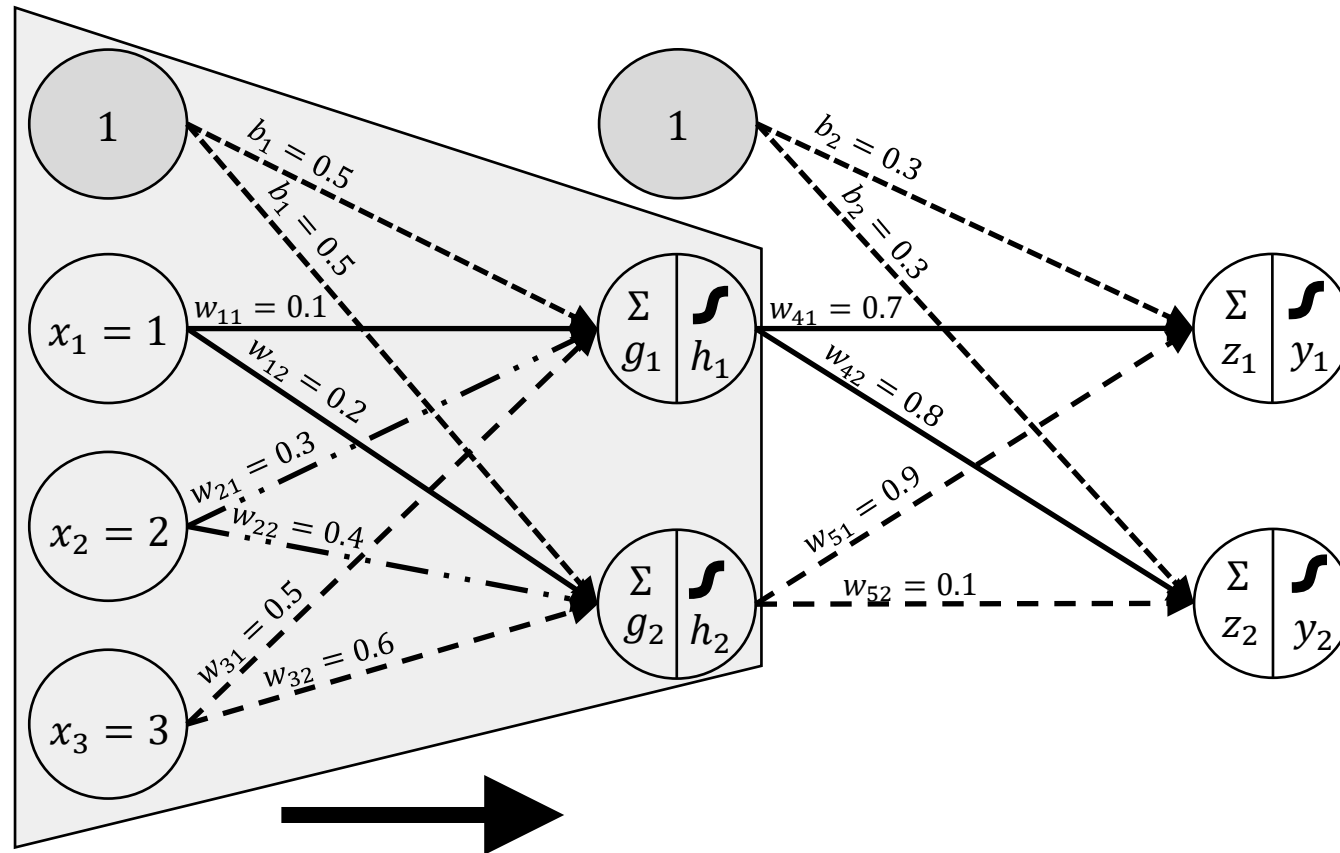
$$\mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1 = \mathbf{g}$$

$$= \begin{pmatrix} 0.1 & 0.3 & 0.5 \\ 0.2 & 0.4 & 0.6 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

$$= \begin{pmatrix} 0.1 \cdot 1 + 0.3 \cdot 2 + 0.5 \cdot 3 \\ 0.2 \cdot 1 + 0.4 \cdot 2 + 0.6 \cdot 3 \end{pmatrix} + \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

$$= \begin{pmatrix} 2.7 \\ 3.3 \end{pmatrix}$$

(2) 순전파를 이용한 출력값 계산



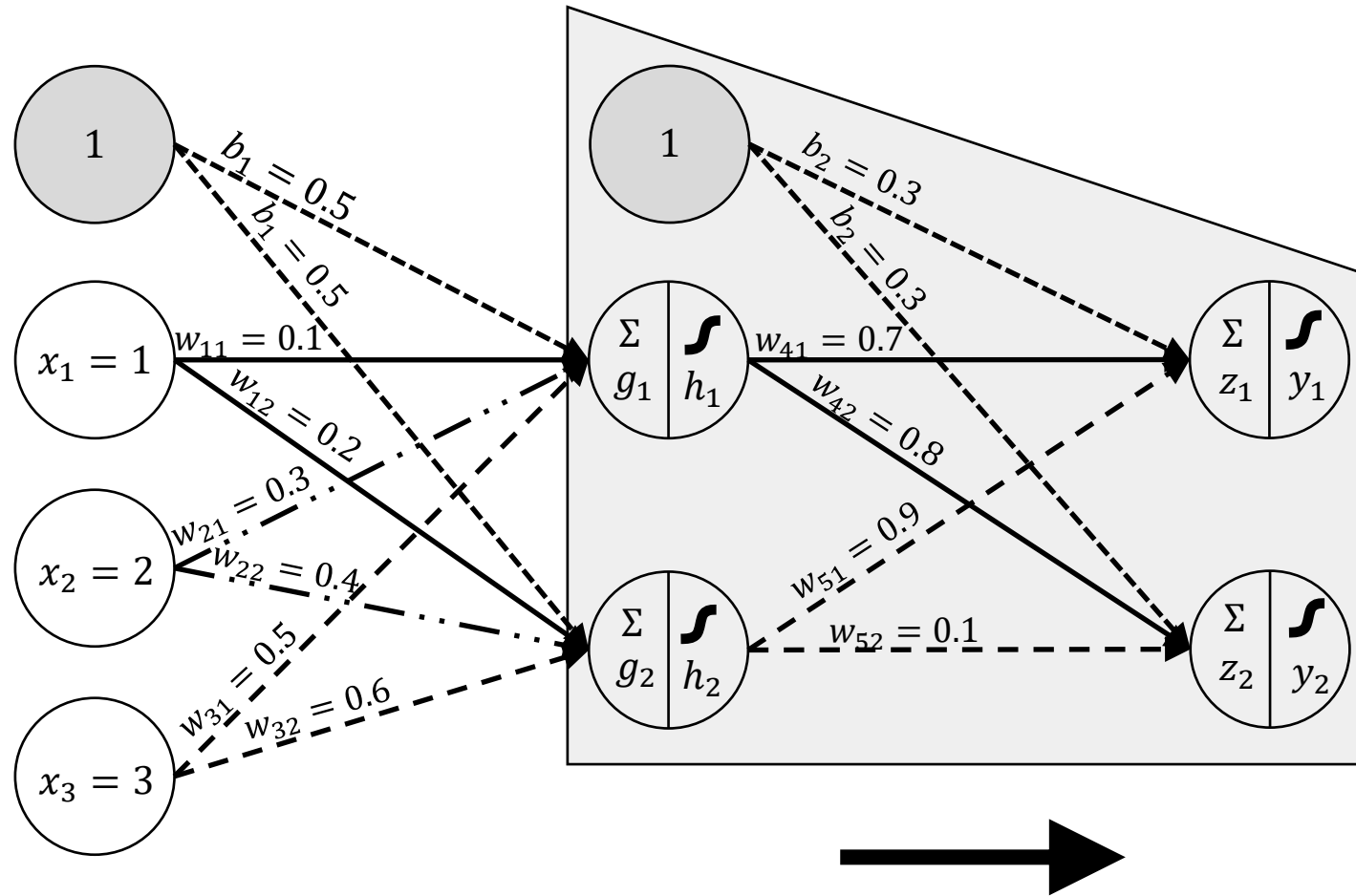
(2) 순전파를 이용한 출력값 계산

활성화 함수 적용

$$h_1 = \phi(g_1) = \phi(2.7) = \frac{1}{1 + \exp(-2.7)} = 0.937$$

$$h_2 = \phi(g_2) = \phi(3.3) = \frac{1}{1 + \exp(-3.3)} = 0.964$$

(2) 순전파를 이용한 출력값 계산



(2) 순전파를 이용한 출력값 계산

$$\mathbf{W}_2^T \mathbf{h} + \mathbf{b}_2 = \mathbf{z}$$

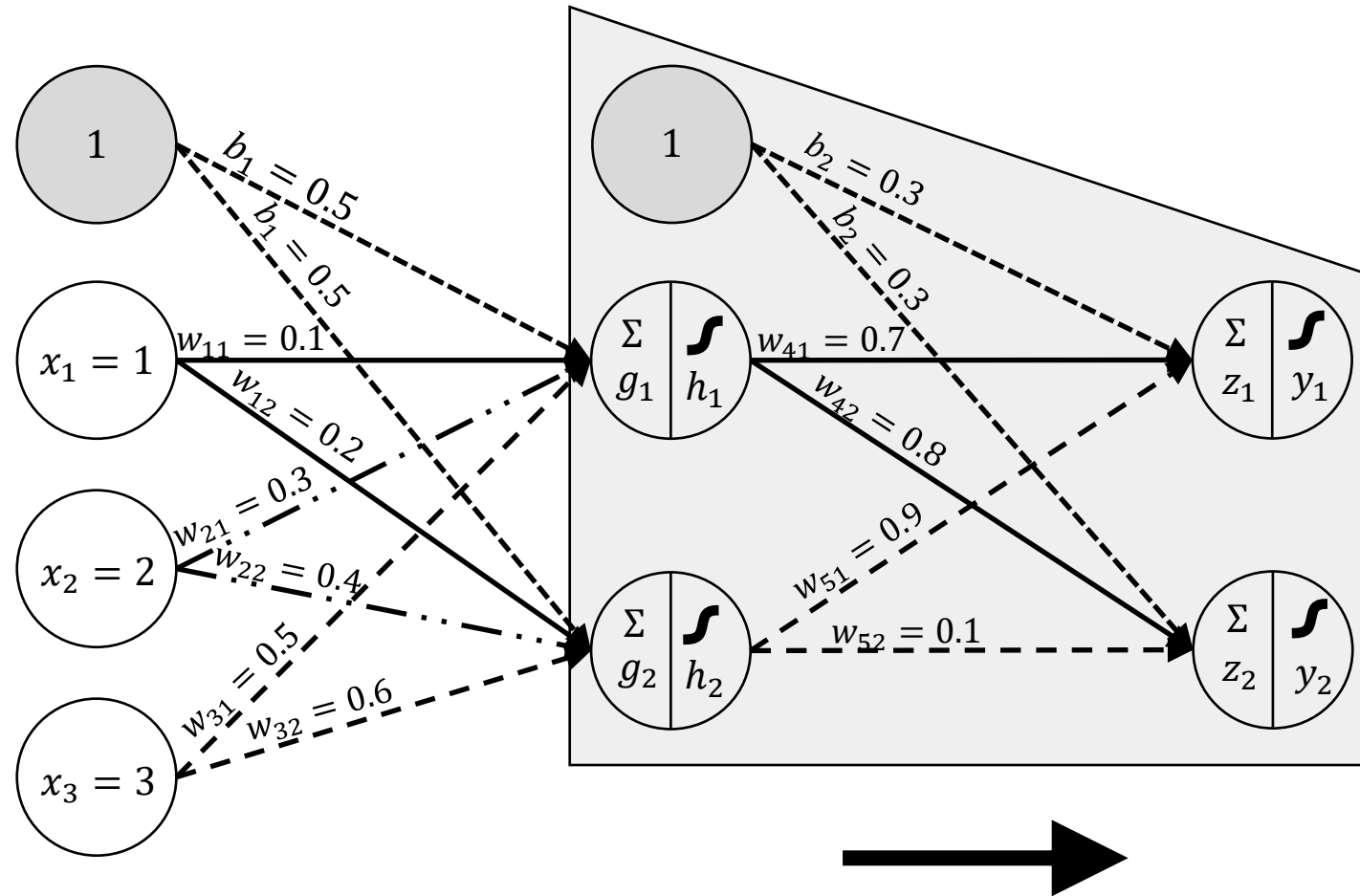
$$\mathbf{W}_2^T \mathbf{h} + \mathbf{b}_2 = \begin{pmatrix} w_{41} & w_{51} \\ w_{42} & w_{52} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} + \begin{pmatrix} b_2 \\ b_2 \end{pmatrix}$$

$$= \begin{pmatrix} 0.7 & 0.9 \\ 0.8 & 0.1 \end{pmatrix} \begin{pmatrix} 0.937 \\ 0.964 \end{pmatrix} + \begin{pmatrix} 0.3 \\ 0.3 \end{pmatrix}$$

$$= \begin{pmatrix} 0.7 \cdot 0.937 + 0.9 \cdot 0.964 \\ 0.8 \cdot 0.937 + 0.1 \cdot 0.964 \end{pmatrix} + \begin{pmatrix} 0.3 \\ 0.3 \end{pmatrix}$$

$$= \begin{pmatrix} 1.823 \\ 1.146 \end{pmatrix}$$

(2) 순전파를 이용한 출력값 계산



(2) 순전파를 이용한 출력값 계산

활성화 함수 적용

$$y_1 = \phi(z_1) = \phi(1.823) = \frac{1}{1 + \exp(-1.823)} = 0.860$$

$$y_2 = \phi(z_2) = \phi(1.146) = \frac{1}{1 + \exp(-1.146)} = 0.758$$

(3) 비용 함수(cost function) 정의 및 1차 미분식 구하기

비용 함수: 오차 제곱합(sum of squared error)

$$C = \frac{1}{2} [(y_1 - t_1)^2 + (y_2 - t_2)^2]$$

$$\frac{\partial C}{\partial y_1} = y_1 - t_1$$

$$\frac{\partial C}{\partial y_2} = y_2 - t_2$$

y_1, y_2 : 신경망 모델을 통해 구한 계산값

t_1, t_2 : 실제값

계산의 편의성을 위해
곱함

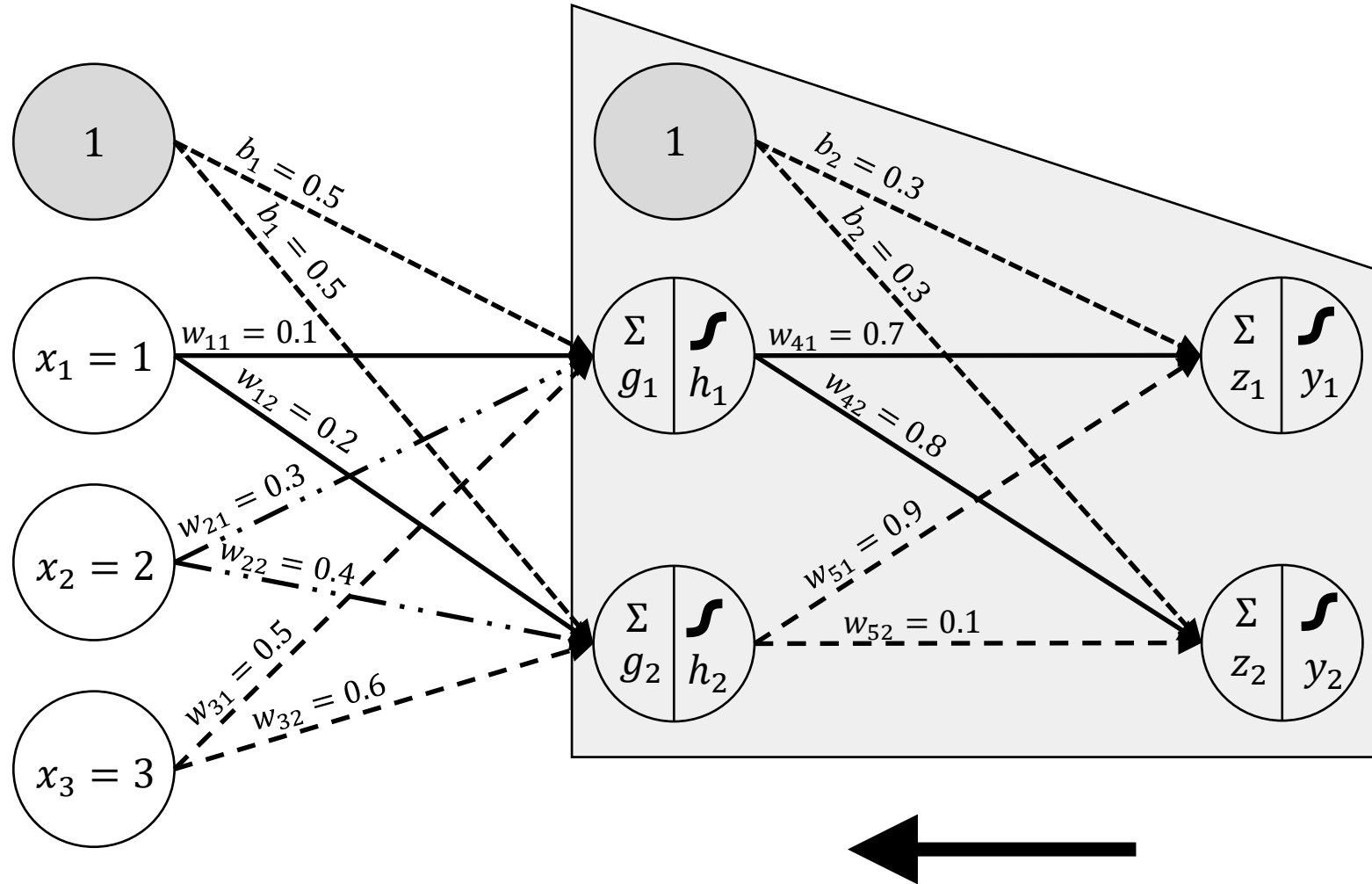
(3) 비용 함수(cost function) 정의 및 1차 미분식 구하기

활성화 함수의 미분값 구하기

$$\phi(x) = \frac{1}{1 + \exp(-x)}$$

$$\frac{d\phi(x)}{dx} = \frac{d}{dx} \left[\frac{1}{1 + \exp(-x)} \right] = \phi(x)(1 - \phi(x))$$

(4) 역전파를 통한 1차 미분값 구하기



(4) 역전파를 통한 1차 미분값 구하기

$$\begin{pmatrix} w_{41} & w_{51} \\ w_{42} & w_{52} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} + \begin{pmatrix} b_2 \\ b_2 \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$$

$$\mathbf{W}_2^T \mathbf{h} + \mathbf{b}_2 = \mathbf{z}$$

$$w_{41}h_1 + w_{51}h_2 + b_2 = z_1$$

$$w_{42}h_1 + w_{52}h_2 + b_2 = z_2$$

$$\frac{\partial \mathbf{z}}{\partial \mathbf{W}_2} = \mathbf{h}, \quad \frac{\partial \mathbf{z}}{\partial \mathbf{b}_2} = 1$$

$$\frac{\partial z_1}{\partial w_{41}} = h_1, \quad \frac{\partial z_1}{\partial w_{51}} = h_2, \quad \frac{\partial z_1}{\partial b_2} = 1$$

$$\frac{\partial z_2}{\partial w_{42}} = h_1, \quad \frac{\partial z_2}{\partial w_{52}} = h_2, \quad \frac{\partial z_2}{\partial b_2} = 1$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial w_{41}} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial w_{41}}$$

$$= (y_1 - t_1) \times [y_1(1 - y_1)] \times h_1$$

$$= (0.860 - 0.5) \times [0.860(1 - 0.860)] \times 0.937$$

$$= 0.360 \times 0.120 \times 0.937 = 0.040$$

$$\frac{\partial C}{\partial w_{42}} = \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial w_{42}}$$

$$= (y_2 - t_2) \times [y_2(1 - y_2)] \times h_1$$

$$= (0.758 - 0.1) \times [0.758(1 - 0.758)] \times 0.937$$

$$= 0.658 \times 0.183 \times 0.937 = 0.113$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial w_{51}} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial w_{51}}$$

$$= (y_1 - t_1) \times [y_1(1 - y_1)] \times h_2$$

$$= (0.860 - 0.5) \times [0.860(1 - 0.860)] \times 0.964$$

$$= 0.360 \times 0.120 \times 0.964 = 0.042$$

$$\frac{\partial C}{\partial w_{52}} = \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial w_{52}}$$

$$= (y_2 - t_2) \times [y_2(1 - y_2)] \times h_2$$

$$= (0.758 - 0.1) \times [0.758(1 - 0.758)] \times 0.964$$

$$= 0.658 \times 0.183 \times 0.964 = 0.116$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial b_2} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial b_2} + \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial b_2}$$

$$= 0.360 \cdot 0.120 \cdot 1 + 0.658 \cdot 0.183 \cdot 1 = 0.164$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial w_{41}} = 0.040$$

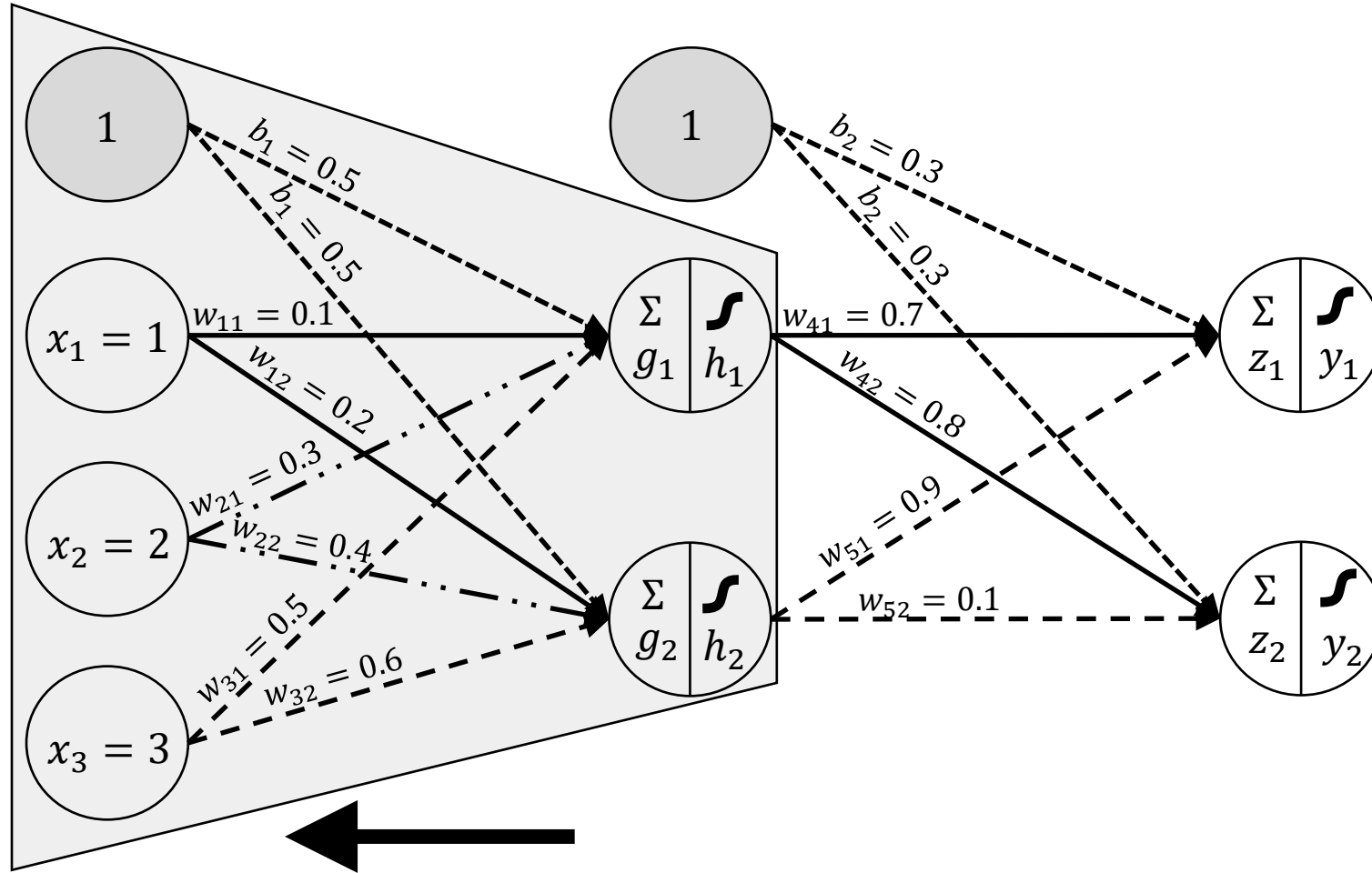
$$\frac{\partial C}{\partial w_{51}} = 0.042$$

$$\frac{\partial C}{\partial b_2} = 0.164$$

$$\frac{\partial C}{\partial w_{42}} = 0.113$$

$$\frac{\partial C}{\partial w_{52}} = 0.116$$

(4) 역전파를 통한 1차 미분값 구하기



(4) 역전파를 통한 1차 미분값 구하기

$$\begin{pmatrix} w_{11} & w_{21} & w_{31} \\ w_{12} & w_{22} & w_{32} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_1 \end{pmatrix}$$

$$\mathbf{W}_1^T \mathbf{x} + \mathbf{b}_1 = \mathbf{g}$$

$$w_{11}x_1 + w_{21}x_2 + w_{31}x_3 + b_1 = g_1$$

$$w_{12}x_1 + w_{22}x_2 + w_{32}x_3 + b_1 = g_2$$

$$\frac{\partial \mathbf{g}}{\partial \mathbf{W}_1} = \mathbf{x}, \quad \frac{\partial \mathbf{g}}{\partial \mathbf{b}_1} = 1$$

$$\frac{\partial g_1}{\partial w_{11}} = x_1, \quad \frac{\partial g_1}{\partial w_{21}} = x_2, \quad \frac{\partial g_1}{\partial w_{31}} = x_3, \quad \frac{\partial g_1}{\partial b_1} = 1$$

$$\frac{\partial g_2}{\partial w_{12}} = x_1, \quad \frac{\partial g_2}{\partial w_{22}} = x_2, \quad \frac{\partial g_2}{\partial w_{32}} = x_3, \quad \frac{\partial g_2}{\partial b_1} = 1$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial h_1} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial h_1} + \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial h_1} = 0.360 \cdot 0.120 \cdot 0.7 + 0.658 \cdot 0.183 \cdot 0.8 = 0.030 + 0.096 = 0.126$$

$$\frac{\partial C}{\partial w_{11}} = \frac{\partial C}{\partial h_1} \frac{\partial h_1}{\partial g_1} \frac{\partial g_1}{\partial w_{11}} = 0.126 \cdot [0.937(1 - 0.937)] \cdot 1 = 0.126 \cdot 0.060 \cdot 1 = 0.008$$

$$\frac{\partial C}{\partial w_{21}} = \frac{\partial C}{\partial h_1} \frac{\partial h_1}{\partial g_1} \frac{\partial g_1}{\partial w_{21}} = 0.126 \cdot [0.937(1 - 0.937)] \cdot 2 = 0.126 \cdot 0.060 \cdot 2 = 0.015$$

$$\frac{\partial C}{\partial w_{31}} = \frac{\partial C}{\partial h_1} \frac{\partial h_1}{\partial g_1} \frac{\partial g_1}{\partial w_{31}} = 0.126 \cdot [0.937(1 - 0.937)] \cdot 3 = 0.126 \cdot 0.060 \cdot 3 = 0.023$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial h_2} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial h_2} + \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial h_2} = 0.360 \cdot 0.120 \cdot 0.9 + 0.658 \cdot 0.183 \cdot 0.1 = 0.039 + 0.012 = 0.051$$

$$\frac{\partial C}{\partial w_{12}} = \frac{\partial C}{\partial h_2} \frac{\partial h_2}{\partial g_2} \frac{\partial g_2}{\partial w_{12}} = 0.051 \cdot [0.964(1 - 0.964)] \cdot 1 = 0.051 \cdot 0.035 \cdot 1 = 0.002$$

$$\frac{\partial C}{\partial w_{22}} = \frac{\partial C}{\partial h_2} \frac{\partial h_2}{\partial g_2} \frac{\partial g_2}{\partial w_{22}} = 0.051 \cdot [0.964(1 - 0.964)] \cdot 2 = 0.051 \cdot 0.035 \cdot 2 = 0.004$$

$$\frac{\partial C}{\partial w_{32}} = \frac{\partial C}{\partial h_2} \frac{\partial h_2}{\partial g_2} \frac{\partial g_2}{\partial w_{32}} = 0.051 \cdot [0.964(1 - 0.964)] \cdot 3 = 0.051 \cdot 0.035 \cdot 3 = 0.005$$

(4) 역전파를 통한 1차 미분값 구하기

$$\frac{\partial C}{\partial b_1} = \frac{\partial C}{\partial y_1} \frac{\partial y_1}{\partial z_1} \frac{\partial z_1}{\partial h_1} \frac{\partial h_1}{\partial g_1} \frac{\partial g_1}{\partial b_1} + \frac{\partial C}{\partial y_2} \frac{\partial y_2}{\partial z_2} \frac{\partial z_2}{\partial h_2} \frac{\partial h_2}{\partial g_2} \frac{\partial g_2}{\partial b_1}$$

$$= 0.360 \cdot 0.120 \cdot 0.7 \cdot 0.060 \cdot 0.5 + 0.658 \cdot 0.183 \cdot 0.1 \cdot 0.035 \cdot 0.5$$

$$= 0.001$$

(5) 파라미터 업데이트

학습률은 $\eta = 0.01$

$$w_{11} \rightarrow w_{11} - \eta \frac{\partial C}{\partial w_{11}} = 0.1 - 0.01 \times 0.008 = 0.09920$$

$$w_{12} \rightarrow w_{12} - \eta \frac{\partial C}{\partial w_{12}} = 0.2 - 0.01 \times 0.002 = 0.19998$$

$$w_{21} \rightarrow w_{21} - \eta \frac{\partial C}{\partial w_{21}} = 0.3 - 0.01 \times 0.015 = 0.29850$$

$$w_{22} \rightarrow w_{22} - \eta \frac{\partial C}{\partial w_{22}} = 0.4 - 0.01 \times 0.004 = 0.39996$$

$$w_{31} \rightarrow w_{31} - \eta \frac{\partial C}{\partial w_{31}} = 0.5 - 0.01 \times 0.023 = 0.49977$$

$$w_{32} \rightarrow w_{32} - \eta \frac{\partial C}{\partial w_{32}} = 0.6 - 0.01 \times 0.005 = 0.59995$$

$$w_{41} \rightarrow w_{41} - \eta \frac{\partial C}{\partial w_{41}} = 0.7 - 0.01 \times 0.040 = 0.69960$$

$$w_{42} \rightarrow w_{42} - \eta \frac{\partial C}{\partial w_{42}} = 0.8 - 0.01 \times 0.113 = 0.79887$$

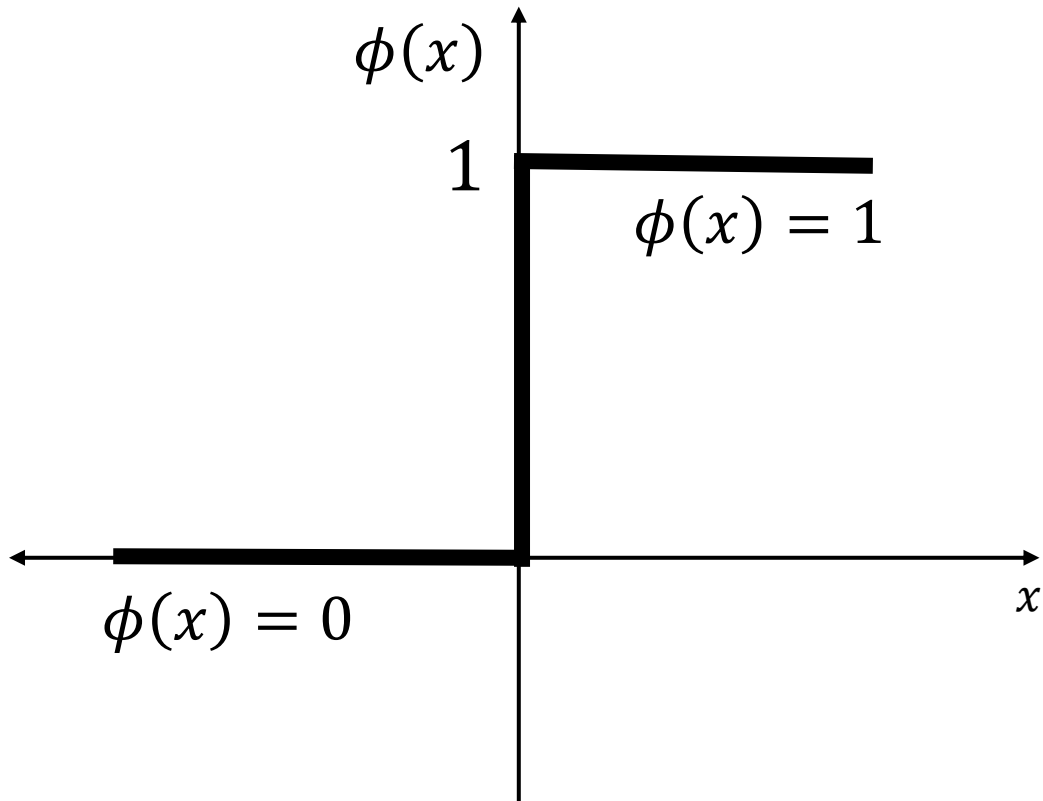
$$w_{51} \rightarrow w_{51} - \eta \frac{\partial C}{\partial w_{51}} = 0.9 - 0.01 \times 0.042 = 0.89958$$

$$w_{52} \rightarrow w_{52} - \eta \frac{\partial C}{\partial w_{52}} = 0.1 - 0.01 \times 0.116 = 0.09884$$

$$b_1 \rightarrow b_1 - \eta \frac{\partial C}{\partial b_1} = 0.5 - 0.01 \times 0.001 = 0.49999$$

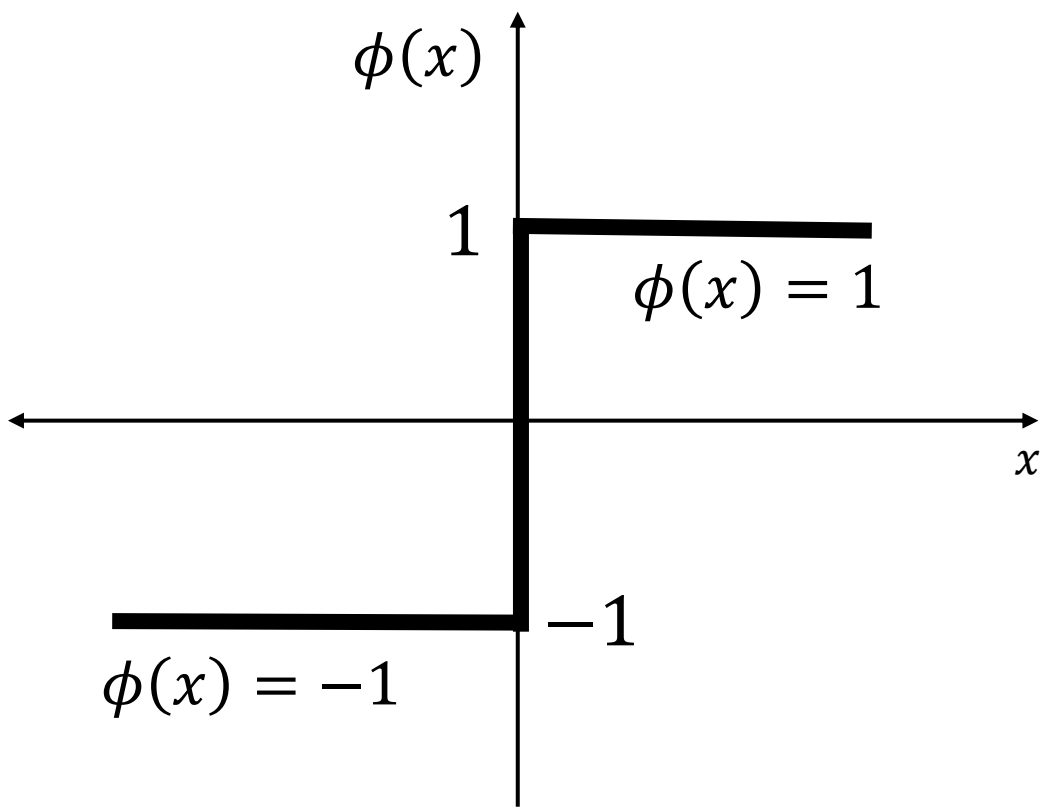
$$b_2 \rightarrow b_2 - \eta \frac{\partial C}{\partial b_2} = 0.1 - 0.01 \times 0.164 = 0.09836$$

활성화 함수 - 계단 함수(step function)



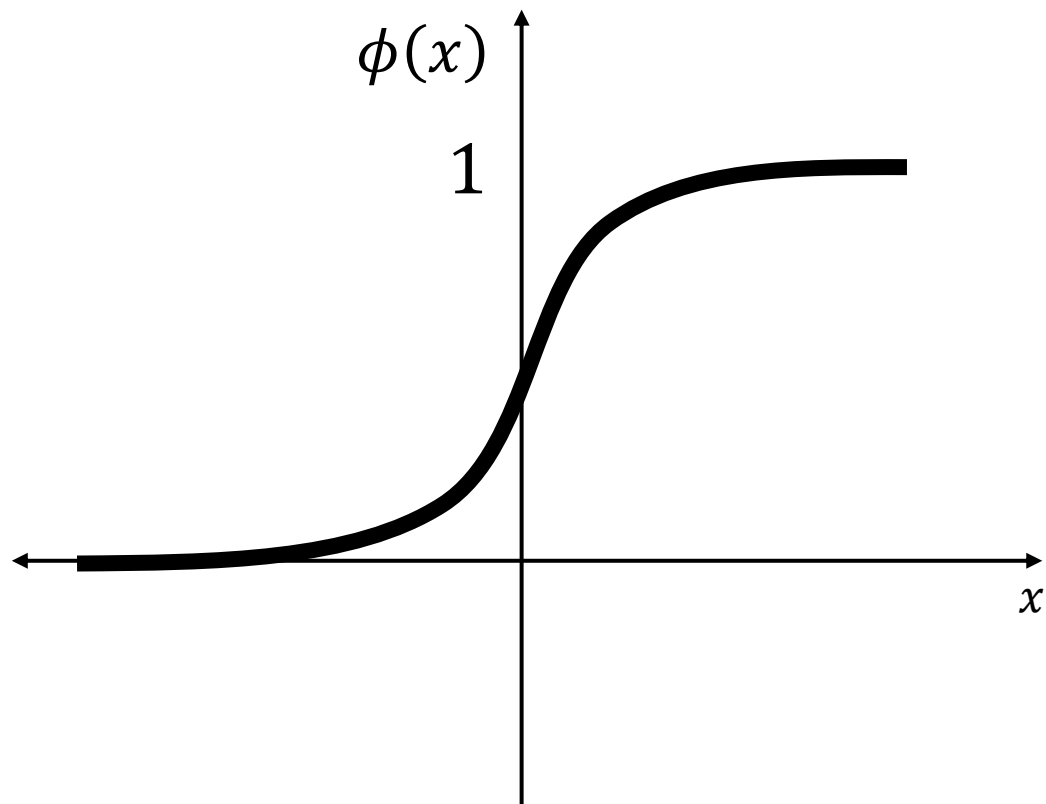
$$\phi(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$

활성화 함수 - 부호 함수(sign function)



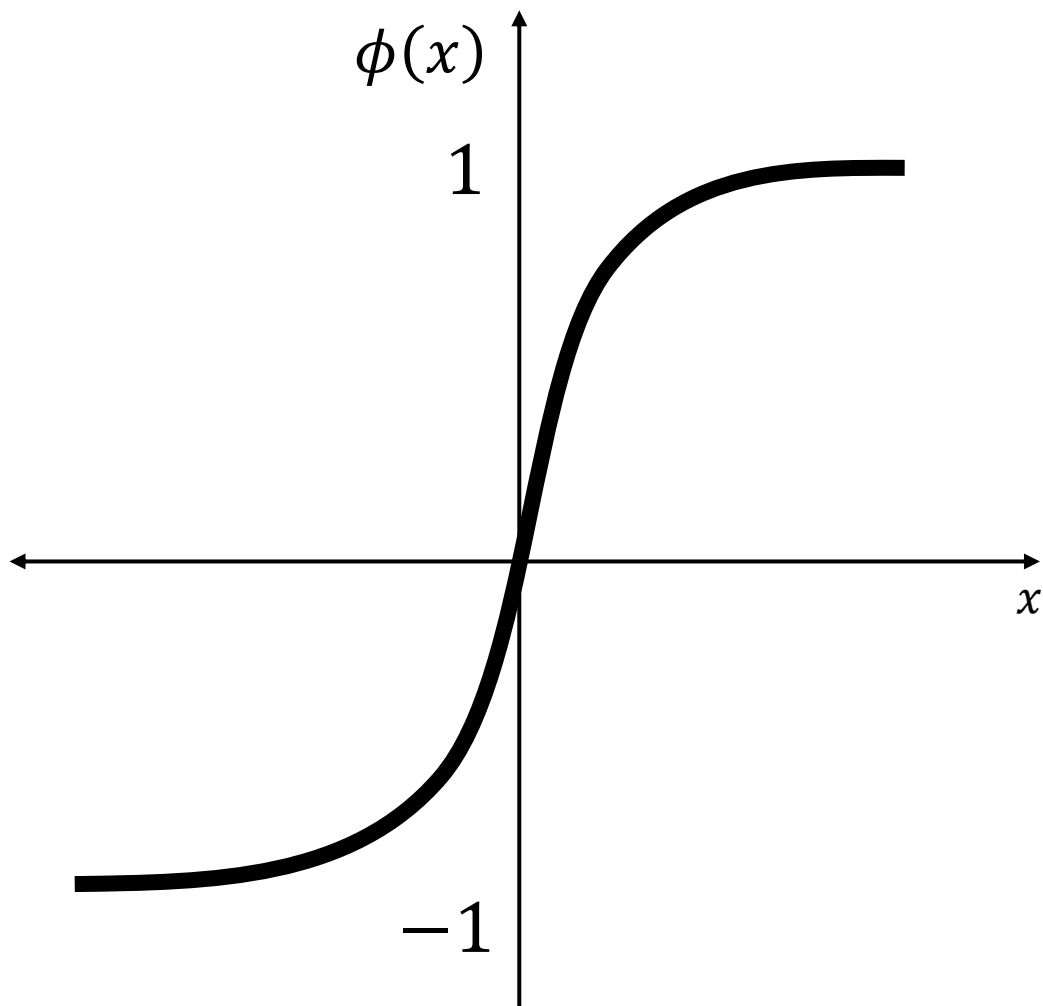
$$\phi(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0 \end{cases}$$

활성화 함수 - 시그모이드 함수(sigmoid function)



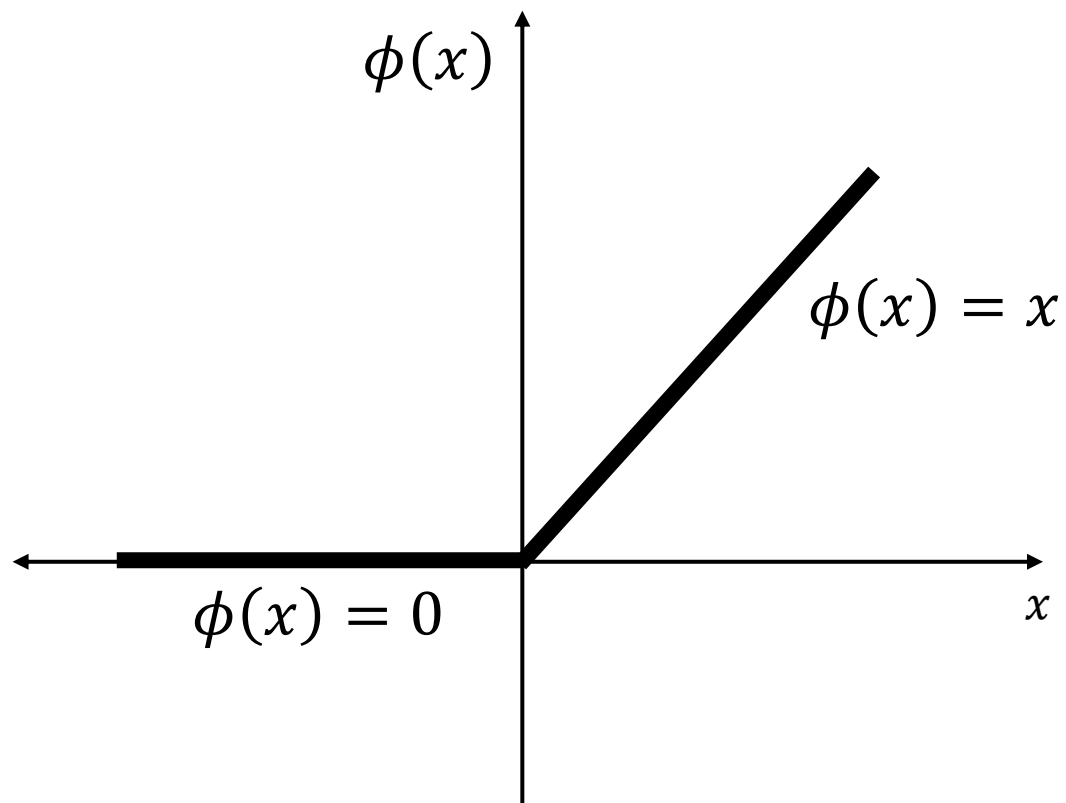
$$\phi(x) = \frac{1}{1 + e^{-x}}$$

활성화 함수 - 하이퍼볼릭 탄젠트(hyperbolic tangent)



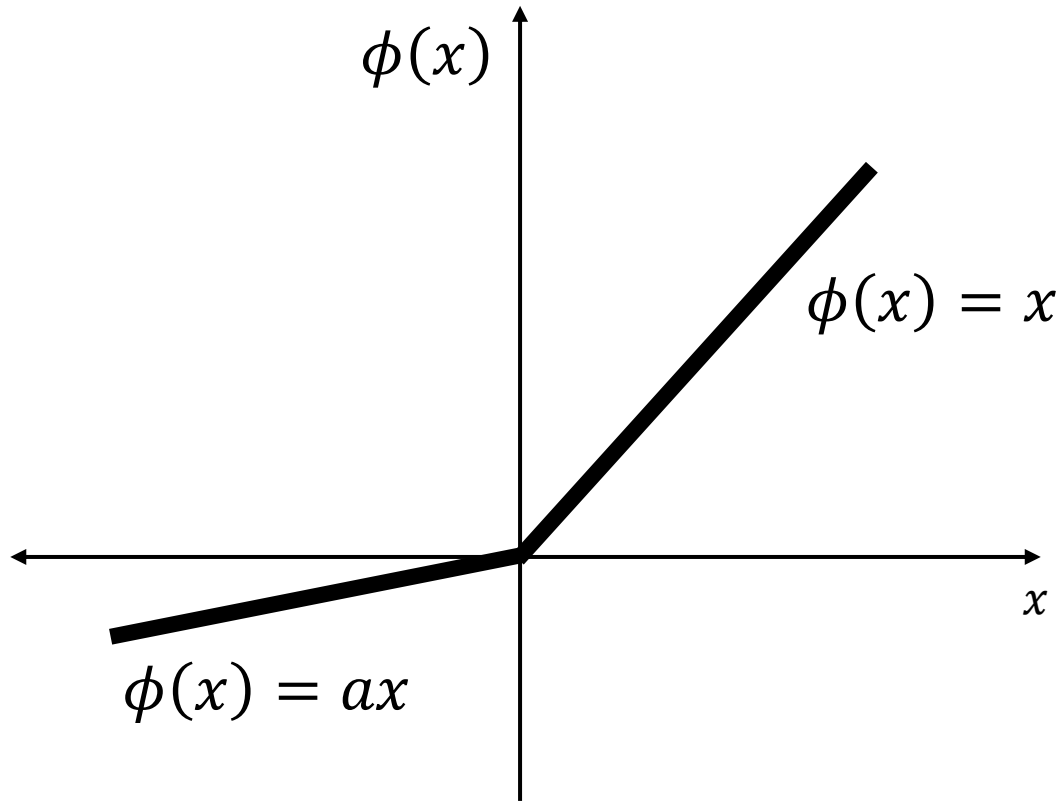
$$\phi(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

활성화 함수 - 렐루 함수(relu function)



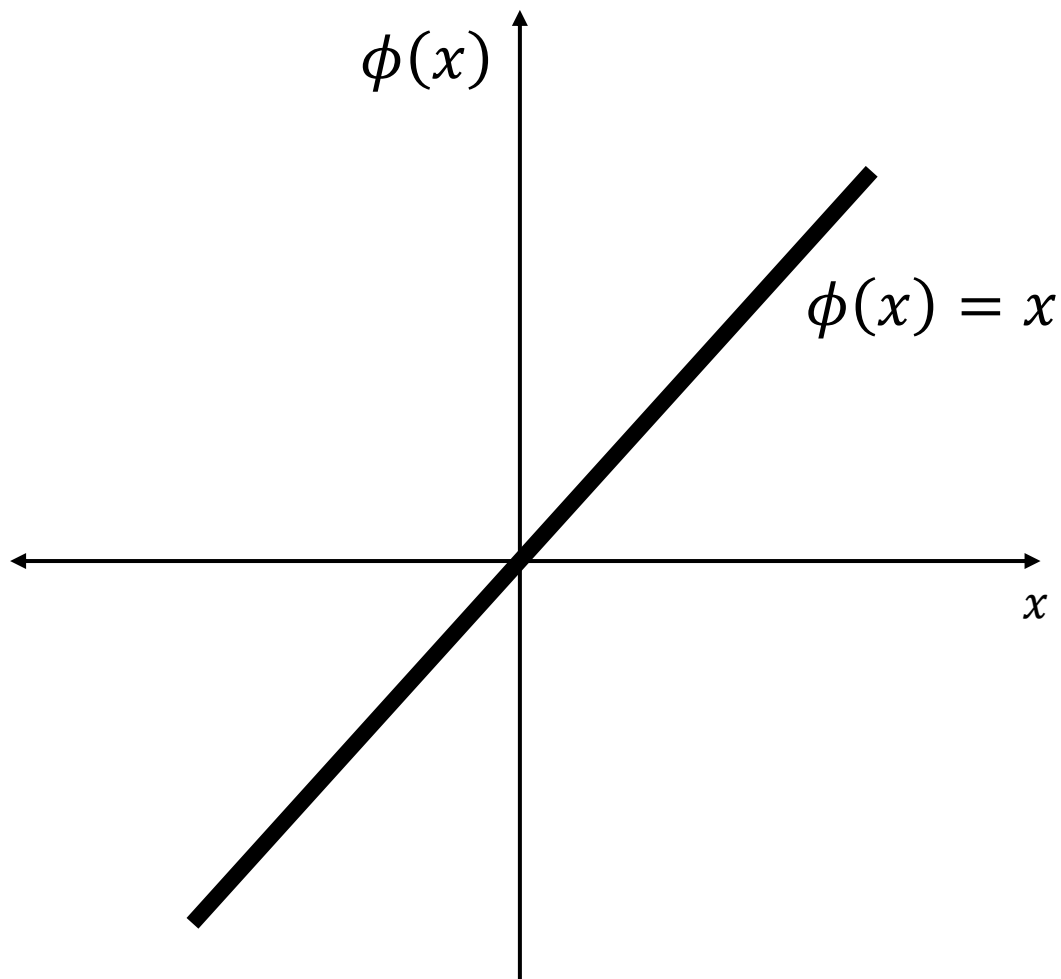
$$\phi(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

활성화 함수 - 리키 렐루(Leaky relu)



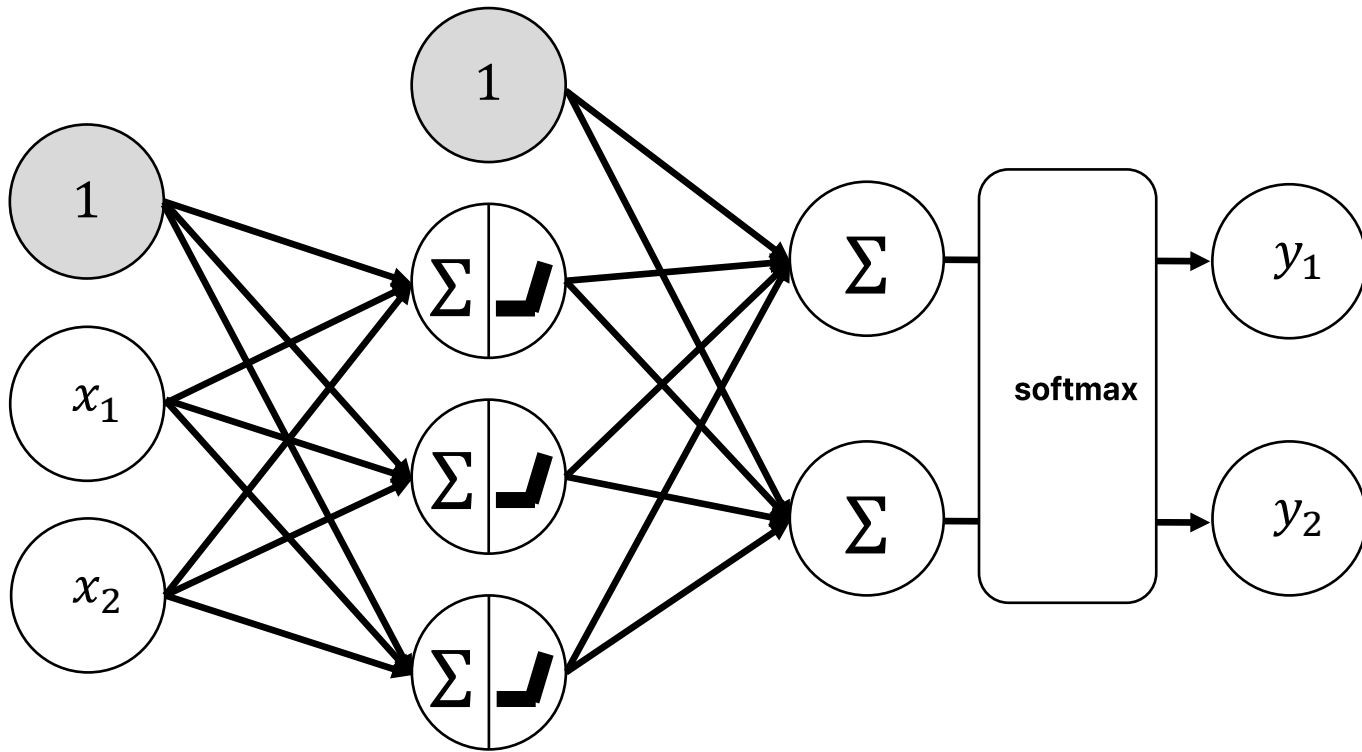
$$\phi(x) = \begin{cases} x, & x > 0 \\ ax, & x \leq 0 \end{cases}$$

활성화 함수 - 항등 함수(identity function)



$$\phi(x) = x$$

활성화 함수 - 소프트맥스 함수(softmax function)



$$\phi(x_k) = \frac{\exp(x_k)}{\sum_{i=1}^n \exp(x_i)}$$

활성화 함수 - 크로스 엔트로피(cross entropy)

피쳐	예측	실제

	예측	실제
포도	0.92	1
레몬	0.05	0
멜론	0.01	0

활성화 함수 - 크로스 엔트로피(cross entropy)

$$0 \leq CrossEntropy(P, Q) < \infty$$

0에 가까울수록 분류를 잘함

$Q(x)$: 추정 분포 $P(x)$: 실제 분포

예측

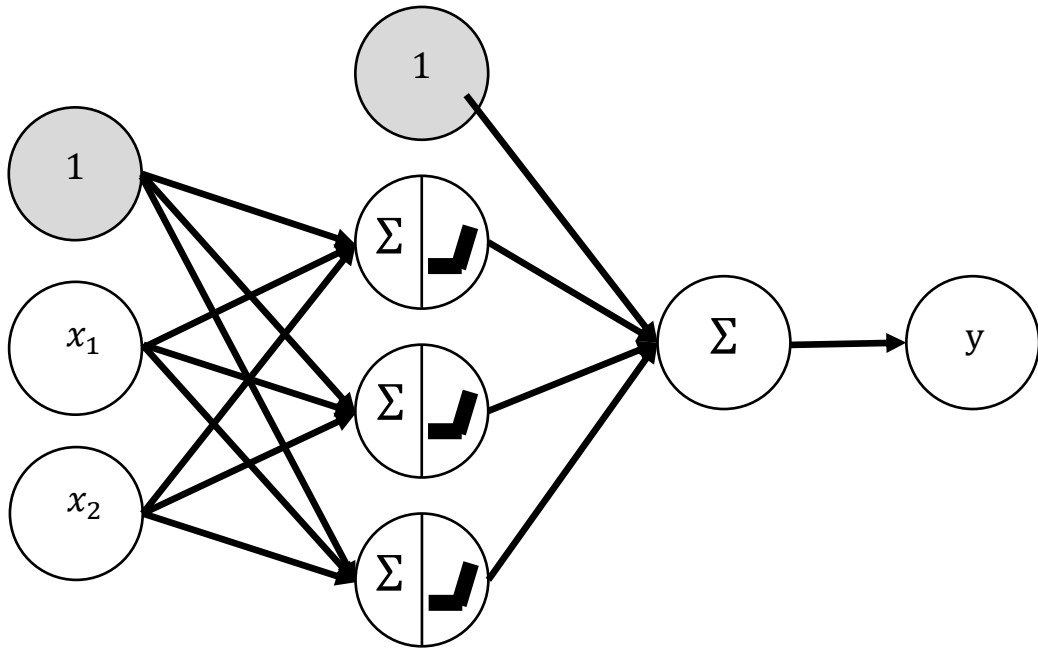
실제

포도	0.92	1
레몬	0.05	0
멜론	0.03	0

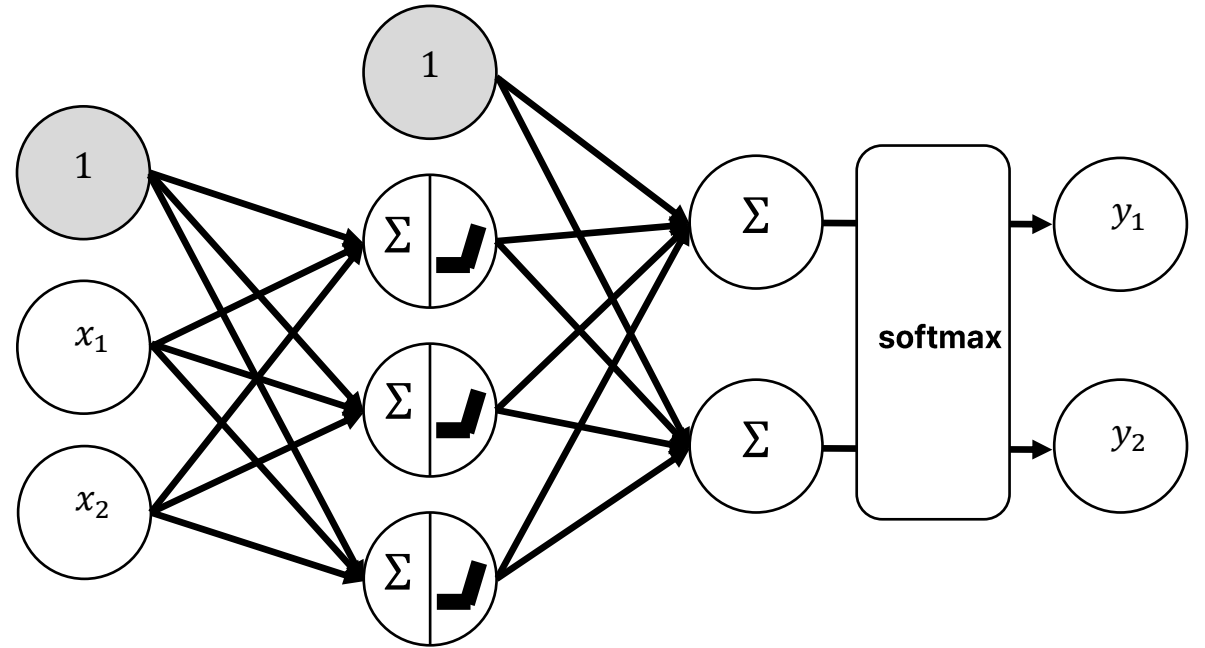
$$CrossEntropy(P, Q) = - \sum P(x) \log Q(x) = -E_P(Q(x))$$

$$= (1 \times \log 0.92 + 0 \times \log 0.05 + 0 \times \log 0.01) = 0.08$$

출력층 활성화 함수 선택

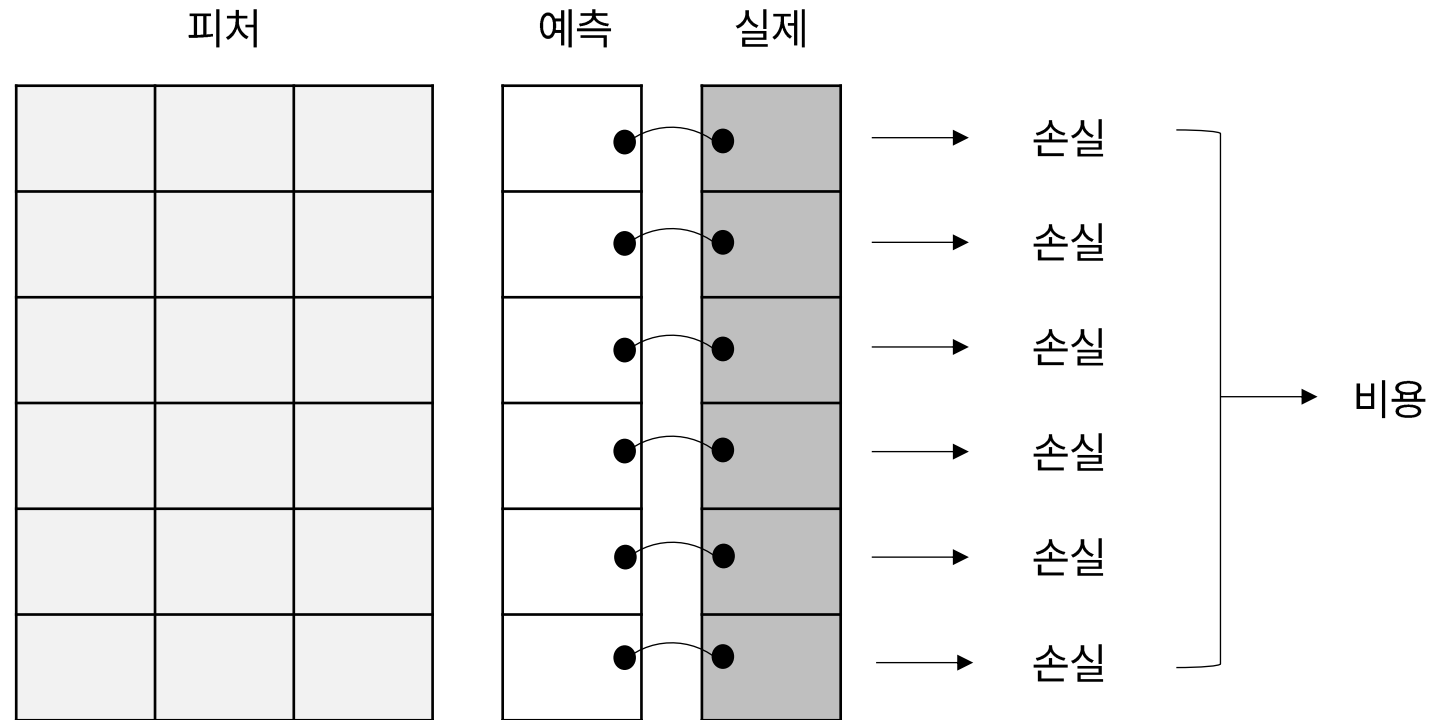


회귀 문제: 출력층에 항등 함수 사용



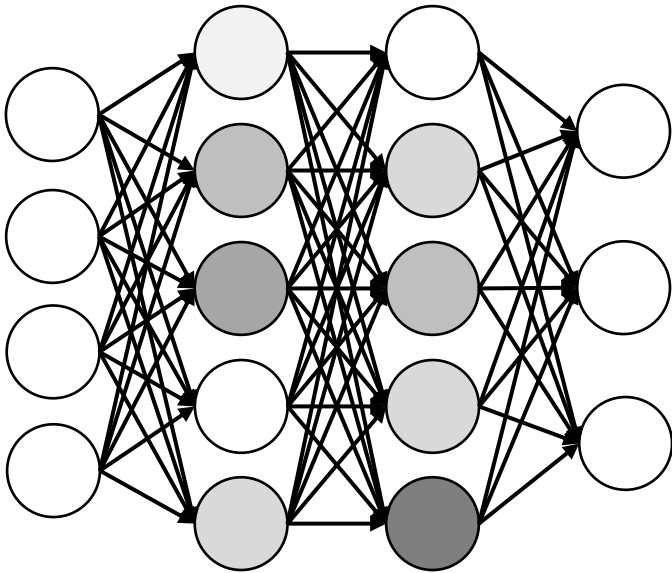
분류 문제: 출력층에 소프트맥스 함수 사용

손실 함수(loss function) vs 비용 함수(cost function)



배치 정규화(batch normalization)

- 해당 층의 분포를 평균 0, 분산 1로 변경 = 평균과 분산을 고정
- 그래디언트 소실문제를 줄임으로써 학습 속도 향상



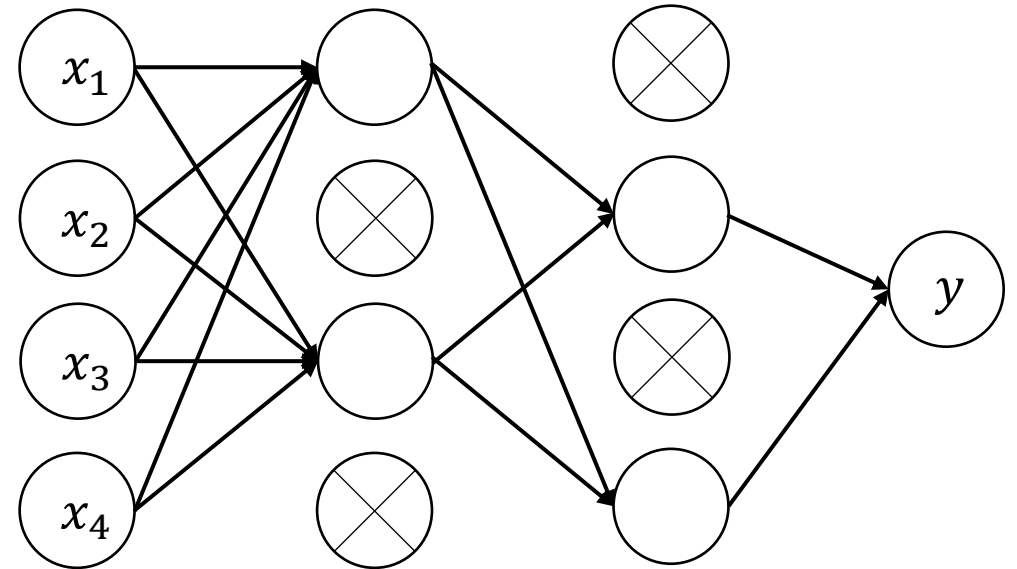
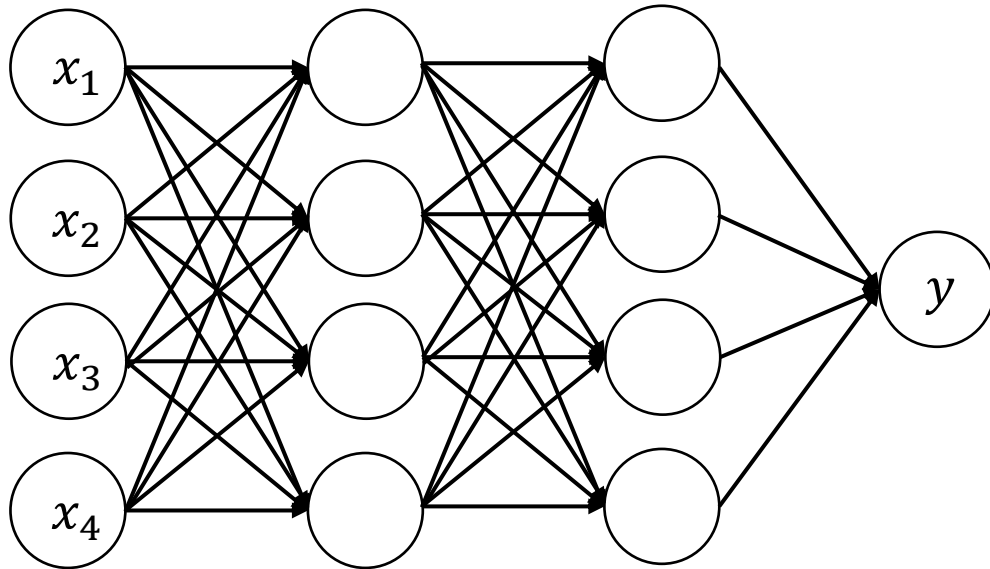
$$\frac{x - \mu}{\sigma}$$

피쳐 1	피쳐 2	피쳐 3	피쳐 4	타겟

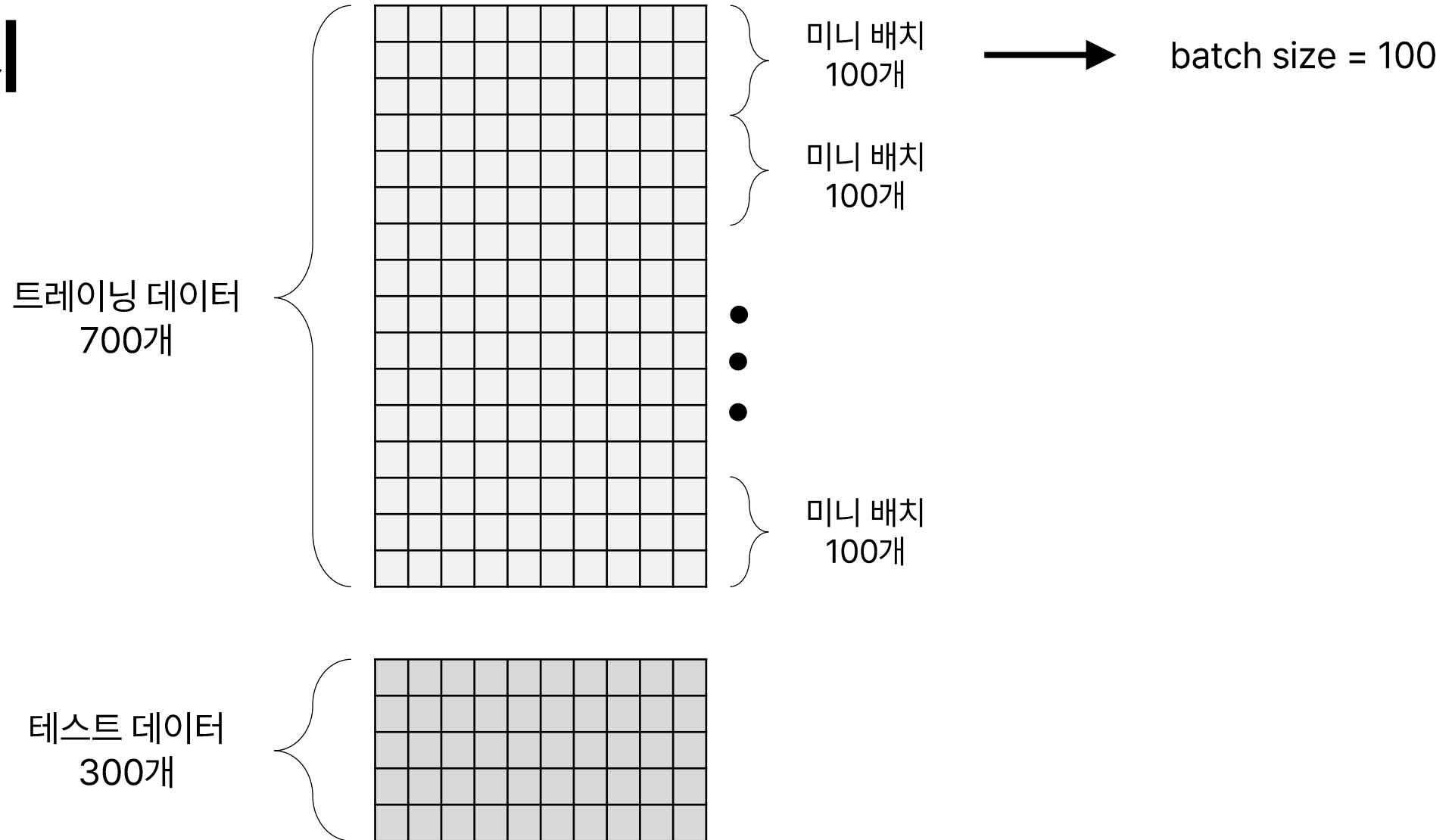
} 미니 배치

드롭 아웃(drop out)

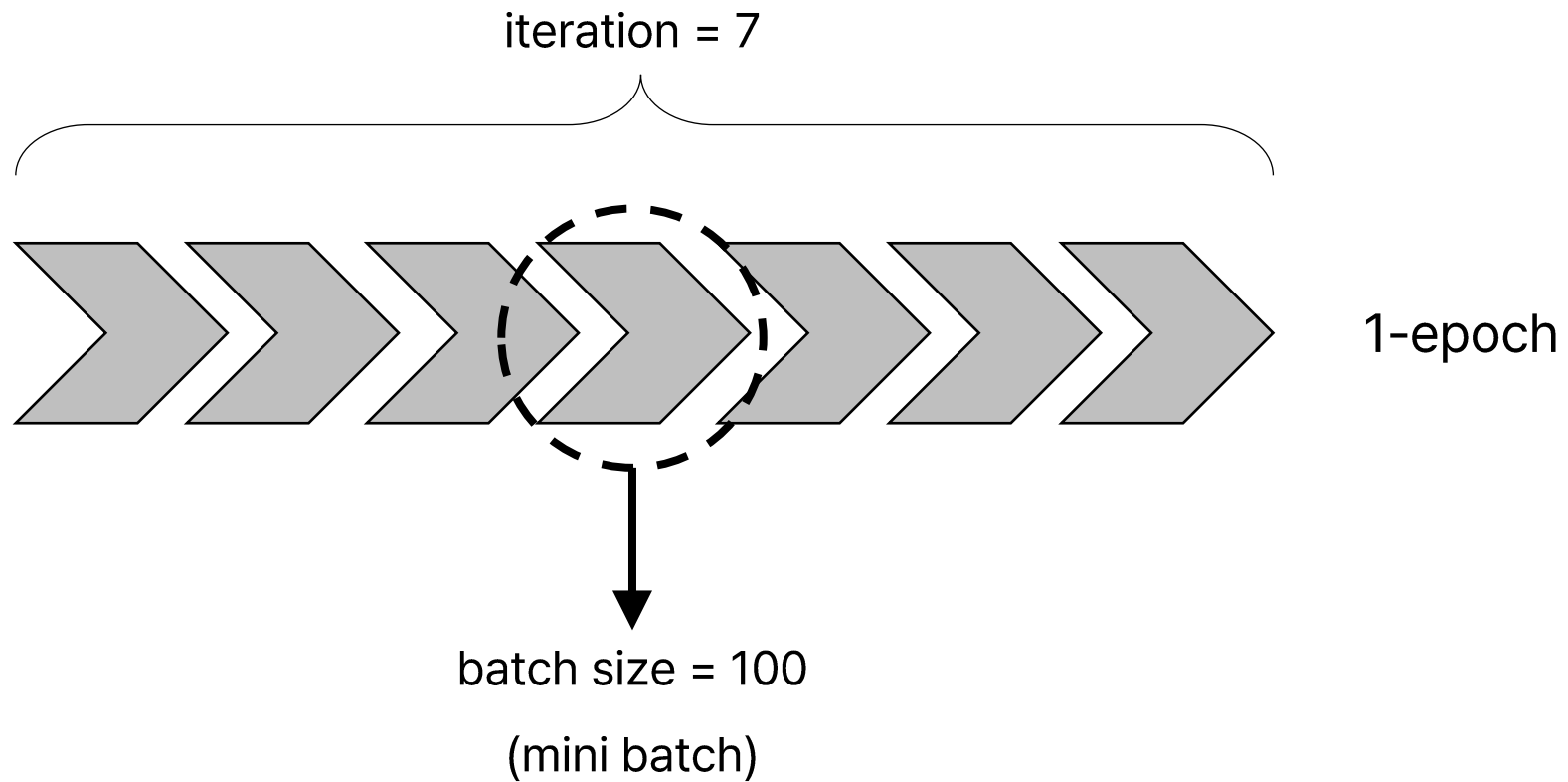
➤ 일부 노드를 사용하지 않음, 오버 피팅 방지



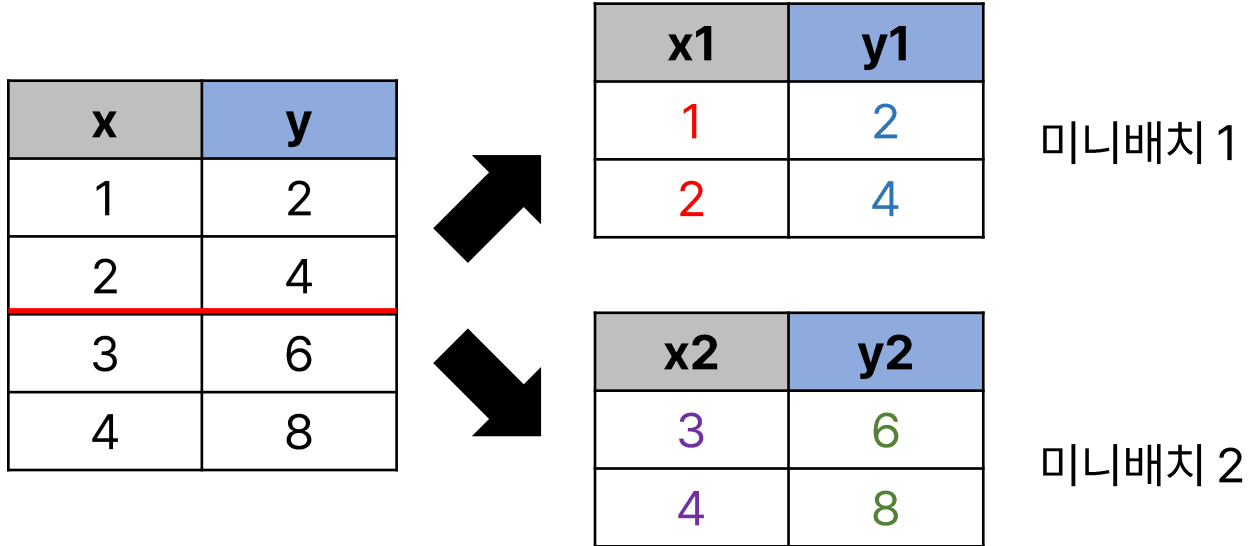
미니 배치



미니 배치



미니 배치



$$\hat{y} = wx$$

- 초기 가중치 $w = 0.5$
- bias b 는 0이라고 가정
- 학습률 $\eta = 0.1$
- 비용함수 = MSE
- 활성화 함수 = 항등 함수

Section 신경망의 개념

미니배치 1)

미니배치 1) 순전파 계산

$$\hat{\mathbf{y}}_1 = \mathbf{w}\mathbf{x}_1 = 0.5 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}$$

미니배치 1) 비용 계산

$$MSE_1 = \frac{1}{2} \sum (\hat{\mathbf{y}}_1 - \mathbf{y}_1)^2 = \frac{1}{2} ((0.5 - 2)^2 + (1 - 4)^2) = 5.625$$

그래디언트 계산

$$\frac{\partial MSE}{\partial \mathbf{w}} = \frac{2}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i)) \approx \frac{1}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i))$$

x1	y1
1	2
2	4

미니배치 1

Section

신경망의 개념

미니배치 1)

미니배치 1) 그래디언트 계산

$$\frac{\partial MSE}{\partial w} = \frac{2}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i)) \approx \frac{1}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i))$$

$$\frac{\partial MSE}{\partial w} = \frac{1}{2} (1 \cdot (0.5 - 2) + 2 \cdot (1 - 4)) = -3.75$$

미니배치 1) 가중치 업데이트

$$w = w - \eta \frac{\partial MSE}{\partial w} = 0.5 - 0.1 \cdot (-3.75) = \mathbf{0.875}$$

x1	y1
1	2
2	4

미니배치 1

Section
신경망의개념

x2	y2
3	6
4	8

미니배치 2

미니배치 2)

미니배치 2) 순전파 계산

$$\widehat{\mathbf{y}}_2 = \mathbf{w}\mathbf{x}_2 = 0.875 \times \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2.625 \\ 3.5 \end{pmatrix}$$

미니배치 1) 비용 계산

$$MSE_1 = \frac{1}{2} \sum (\widehat{\mathbf{y}}_2 - \mathbf{y}_2)^2 = \frac{1}{2} ((2.625 - 6)^2 + (3.5 - 8)^2) = 15.82$$

그레디언트 계산

$$\frac{\partial MSE}{\partial \mathbf{w}} = \frac{2}{n} \sum_{i=1}^n (x_i(\widehat{y}_i - y_i)) \approx \frac{1}{n} \sum_{i=1}^n (x_i(\widehat{y}_i - y_i))$$

Section 신경망의개념

미니배치 2)

미니배치 2) 그레디언트 계산

$$\frac{\partial MSE}{\partial w} = \frac{2}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i)) \approx \frac{1}{n} \sum_{i=1}^n (x_i(\hat{y}_i - y_i))$$

$$\frac{\partial MSE}{\partial w} = \frac{1}{2} (3 \cdot (2.625 - 6) + 4 \cdot (3.5 - 8)) = -14.06$$

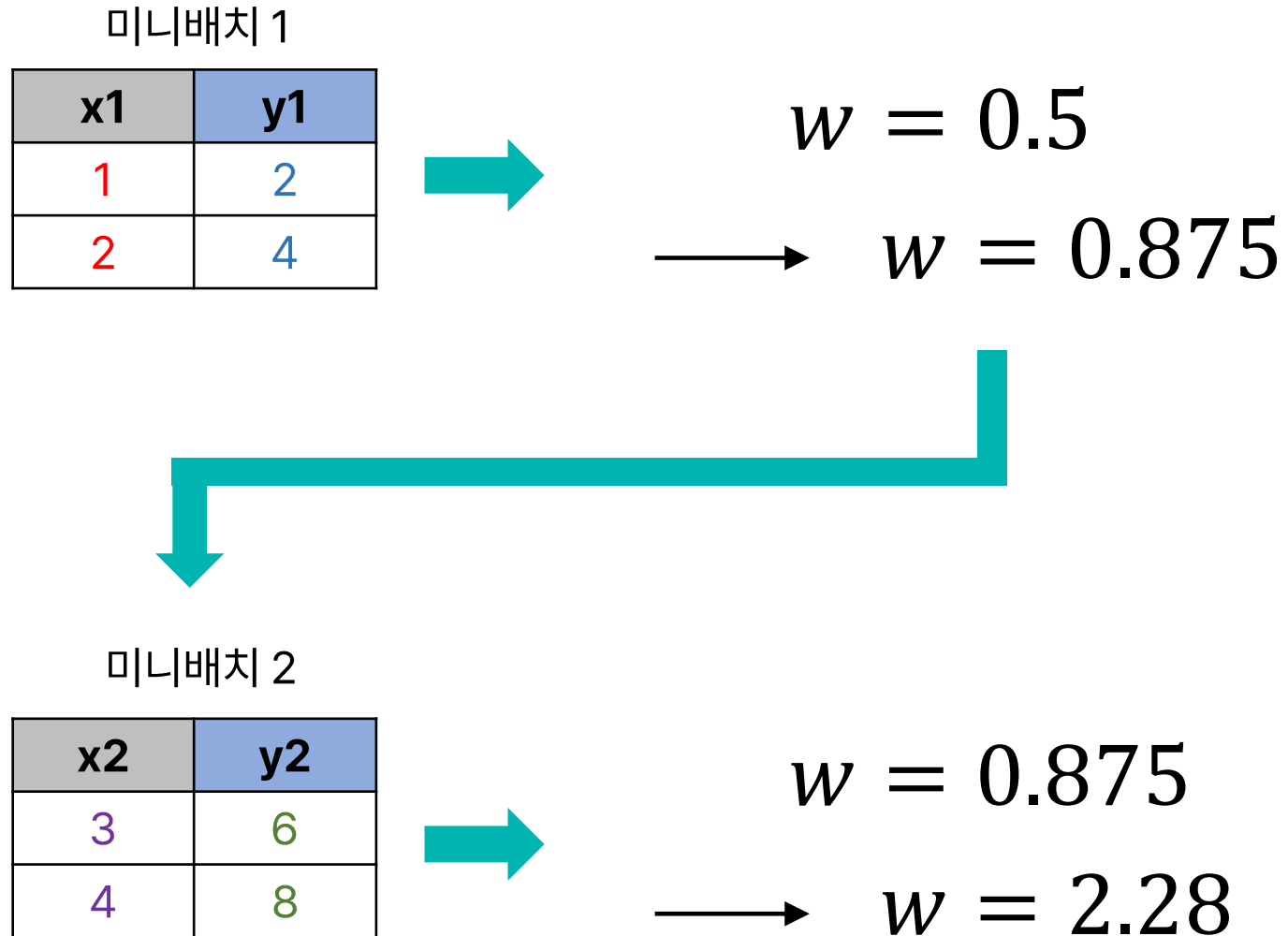
미니배치 2) 가중치 업데이트

$$w = w - \eta \frac{\partial MSE}{\partial w} = 0.875 - 0.1 \cdot (-14.0625) = \mathbf{2.28}$$

x2	y2
3	6
4	8

미니배치 2

미니 배치



감사합니다.

Q & A